DOT OR SCALAR PRODUCT OF VECTORS

SYNOPSIS

- Let $\bar{a}$ and $\bar{b}$ be two nonzero vectors and θ be the angle between them, then the scalar product or dot product of $\bar{a}$ and $\bar{b}$ denoted by $\bar{a}.\bar{b}$ is defined as $\bar{a}.\bar{b} = |\bar{a}|\,|\bar{b}|\cos\theta$
- If $(\bar{a},\bar{b}) = \theta$, then $\cos\theta = \dfrac{\bar{a}.\bar{b}}{|\bar{a}||\bar{b}|}$
- Let $\bar{i}, \bar{j}, \bar{k}$ be unit orthogonal vectors i.e $|\bar{i}| = |\bar{j}| = |\bar{k}| = 1$ and $(\bar{i},\bar{j}) = (\bar{j},\bar{k}) = (\bar{k},\bar{i}) = \dfrac{\pi}{2}$,

  then $\bar{i}\cdot\bar{i} = \bar{j}\cdot\bar{j} = \bar{k}\cdot\bar{k} = 1$

  $\bar{i}\cdot\bar{j} = \bar{j}\cdot\bar{k} = \bar{k}\cdot\bar{i} = 0$

  $\bar{j}\cdot\bar{i} = \bar{k}\cdot\bar{j} = \bar{i}\cdot\bar{k} = 0$
- If $\bar{a} = a_1\bar{i} + a_2\bar{j} + a_3\bar{k}, \bar{b} = b_1\bar{i} + b_2\bar{j} + b_3\bar{k}$ Then

  (i) $\bar{a}.\bar{b} = a_1b_1 + a_2b_2 + a_3b_3$

  (ii) $\bar{a}.\bar{a} = a_1^2 + a_2^2 + a_3^2$ (iii) $\bar{a}.\bar{a} = |\bar{a}|^2$
- Let $\bar{a} = a_1\bar{i} + a_2\bar{j} + a_3\bar{k}$, $\bar{b} = b_1\bar{i} + b_2\bar{j} + b_3\bar{k}$ and let $(\bar{a},\bar{b}) = \theta$ then

  $$\cos\theta = \frac{a_1b_1 + a_2b_2 + a_3b_3}{\sqrt{a_1^2 + a_2^2 + a_3^2}\sqrt{b_1^2 + b_2^2 + b_3^2}}$$
- Let $(\bar{a},\bar{b}) = \theta$, then

  (i) $\theta < 90° \Leftrightarrow \bar{a}.\bar{b} > 0$ (ii) $\theta > 90^0 \Leftrightarrow \bar{a}.\bar{b} < 0$

  (iii) $\theta = 90^0 \Leftrightarrow \bar{a}.\bar{b} = 0$ ($\bar{a}$ is perpendicular to $\bar{b}$)

W.E-1: *If $\bar{a} = \bar{i} - \lambda\bar{j} + 2\bar{k}$ and $\bar{b} = 8\bar{i} + 6\bar{j} - \bar{k}$ are perpendicular then λ is*

Sol: Let $\bar{a} = \bar{i} - \lambda\bar{j} + 2\bar{k}$ and $\bar{b} = 8\bar{i} + 6\bar{j} - \bar{k}$ Since, $\bar{a},\bar{b}$ are at right angles, $\bar{a}.\bar{b} = 0$

$\Rightarrow (\bar{i} - \lambda\bar{j} + 2\bar{k}).(8\bar{i} + 6\bar{j} - \bar{k}) = 0$

$\Rightarrow 8 - 6\lambda - 2 = 0 \Rightarrow \lambda = 1$

- If $\bar{a}$ and $\bar{b}$ are parallel and are in the same direction ($\bar{a},\bar{b}$ are like vectors) then $\bar{a}.\bar{b} = |\bar{a}||\bar{b}|$
- If $\bar{a}$ and $\bar{b}$ are parallel and are in the opposite direction ($\bar{a},\bar{b}$ are unlike vectors) then $\bar{a}.\bar{b} = -|\bar{a}||\bar{b}|$
- Let $\bar{a},\bar{b}$ are two unit vectors and $\bar{a}.\bar{b} = \cos\theta$
- If $\bar{a},\bar{b}$ are any two vectors, then $\bar{a}.\bar{b} = 0$ either $\bar{a} = \bar{0}$ (or) $\bar{b} = \bar{0}$ (or) $\bar{a} \perp \bar{b}$. However if $\bar{a},\bar{b}$ are two non-zero perpendicular vectors, then $\bar{a}.\bar{b} = 0$
- **Geometrical interpretation of scalar product :**

  Let $\bar{a}$ and $\bar{b}$ be two vectors represented by $\overline{OA}$ and $\overline{OB}$ respectively. Let θ be the angle between $\overline{OA}$ and $\overline{OB}$. Draw $\overline{BL} \perp \overline{OA}$ and $\overline{AM} \perp \overline{OB}$. From triangle OBL and OAM, We have $OL = OB\cos\theta$ and $OM = OA\cos\theta$. Here OL and OM are known as projections of $\bar{b}$ on $\bar{a}$ and $\bar{a}$ on $\bar{b}$.

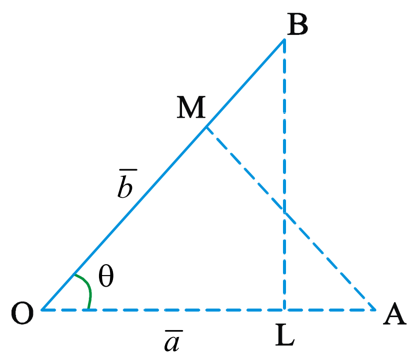
  $\bar{a}.\bar{b} = |\bar{a}||\bar{b}|\cos\theta = |\bar{a}|(OB\cos\theta) = |\bar{a}|(OL)$

  = (magnitude of $\bar{a}$) (projection of $\bar{b}$ on $\bar{a}$)

  again $\bar{a}.\bar{b} = |\bar{a}||\bar{b}|\cos\theta = |\bar{b}|(|\bar{a}|\cos\theta)$

  $= |\bar{b}|(OA\cos\theta) = |\bar{b}|(OM)$

  = (magnitude of $\bar{b}$) (projection of $\bar{a}$ *on* $\bar{b}$)



- Let $\bar{a}$ and $\bar{b}$ be two nonzero vectors. Then Component vector of $\bar{b}$ on $\bar{a}$ (or) orthogonal projection of $\bar{b}$ on $\bar{a}$ is $\frac{(\bar{b}.\bar{a})\bar{a}}{|\bar{a}|^2}$

WE-2: Orthogonal projection of $\bar{b} = 2\bar{i} - 3\bar{j} + 6\bar{k}$ ***on*** $\bar{a} = \bar{i} + 2\bar{j} + 2\bar{k}$ ***is***

Sol: Orthogonal projection

$$= \frac{[(2\bar{i} - 3\bar{j} + 6\bar{k}).(\bar{i} + 2\bar{j} + 2\bar{k})](\bar{i} + 2\bar{j} + 2\bar{k})}{|\bar{i} + 2\bar{j} + 2\bar{k}|^2}$$

$$= \frac{(2-6+12)}{9}(\bar{i} + 2\bar{j} + 2\bar{k}) = \frac{8}{9}(\bar{i} + 2\bar{j} + 2\bar{k})$$

- Component vector of $\bar{a}$ on $\bar{b}$ (or) orthogonal projection of $\bar{a}$ on $\bar{b}$ is $\frac{(\bar{a}.\bar{b})\bar{b}}{|\bar{b}|^2}$

WE-3: If $\bar{a} = 2\bar{i} + \bar{j} + 2\bar{k}, \bar{b} = 5\bar{i} - 3\bar{j} + \bar{k}$ ***then orthogonal projection of*** $\bar{a}$ ***on*** $\bar{b}$ ***is***

Sol: Orthogonal projection of $\bar{a}$ on $\bar{b}$ is $\frac{(\bar{a}.\bar{b})\bar{b}}{|\bar{b}|^2} =$

$$= \frac{(10-3+2)(5\bar{i} - 3\bar{j} + \bar{k})}{25+9+1} = \frac{9(5\bar{i} - 3\bar{j} + \bar{k})}{35}$$

- The orthogonal projection of $\bar{b}$ in the direction perpendicular to that of $\bar{a}$ is $\bar{b} - \frac{(\bar{b}.\bar{a})\bar{a}}{|\bar{a}|^2}$

WE-4: The orthogonal projection of $\bar{b} = 3\bar{i} + 2\bar{j} - 5\bar{k}$ ***on a vector perpendicular to*** $\bar{a} = 2\bar{i} - \bar{j} + 2\bar{k}$ ***is***

Sol: Orthogonal projection of $\bar{b}$ *on* $\bar{a}$

$$= 3\bar{i} + 2\bar{j} - 5\bar{k} - \frac{(3\bar{i} + 2\bar{j} - 5\bar{k}).(2\bar{i} - \bar{j} + 2\bar{k})}{4+1+4}(2\bar{i} - \bar{j} + 2\bar{k})$$

$$= 3\bar{i} + 2\bar{j} - 5\bar{k} - \frac{6-2-10}{9}(2\bar{i} - \bar{j} + 2\bar{k})$$

$$= (3\bar{i} + 2\bar{j} - 5\bar{k}) + \frac{2}{3}(2\bar{i} - \bar{j} + 2\bar{k})$$

$$= \frac{13\bar{i} + 4\bar{j} - 11\bar{k}}{3}$$

- The length of the orthogonal projection of $\bar{b}$ on $\bar{a}$ is $\left|\frac{(\bar{a}.\bar{b})}{|\bar{a}|}\right|$
- The length of the orthogonal projection of $\bar{a}$ on $\bar{b}$ is $\left|\frac{(\bar{a}.\bar{b})}{|\bar{b}|}\right|$

WE-5: The length of orthogonal projection of $\bar{a} = 2\bar{i} - 3\bar{j} + \bar{k}$ ***on*** $\bar{b} = 4\bar{i} - 4\bar{j} + 7\bar{k}$ ***is***

Sol: The length of the orthogonal projection of $\bar{a}$ on $\bar{b}$ is $\left|\frac{(\bar{a}.\bar{b})}{|\bar{b}|}\right| = \frac{|8+12+7|}{\sqrt{16+16+49}} = \frac{27}{9} = 3$

- The scalar product is commutative i.e., $\bar{a}\cdot\bar{b} = \bar{b}\cdot\bar{a}$
- The scalar product is distributive over vector addition i.e., $\bar{a}\cdot(\bar{b}+\bar{c}) = \bar{a}\cdot\bar{b} + \bar{a}\cdot\bar{c}, (\bar{b}+\bar{c})\cdot\bar{a} = \bar{b}\cdot\bar{a} + \bar{c}\cdot\bar{a}$
- $(l\bar{a}).\bar{b} = \bar{a}.(l\bar{b}) = l(\bar{a}.\bar{b})$ where l is a scalar
- $\bar{a}.\bar{a} \ge 0; |\bar{a}.\bar{b}| \le |\bar{a}||\bar{b}|; |\bar{a} - \bar{b}| \ge ||\bar{a}| - |\bar{b}||$

  $|\bar{a} + \bar{b}| \le |\bar{a}| + |\bar{b}|$
- Cauchy schwartz in equality : Let a_1, a_2, a_3 *and* b_1, b_2, b_3 be real numbers. Then $(a_1b_1 + a_2b_2 + a_3b_3)^2 \le (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2)$ and equality holds If $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3}$
- $|\bar{a} + \bar{b}|^2 = |\bar{a}|^2 + |\bar{b}|^2 + 2\bar{a}.\bar{b}$
- $|\bar{a} - \bar{b}|^2 = |\bar{a}|^2 + |\bar{b}|^2 - 2\bar{a}.\bar{b}$
- $|\bar{a} + \bar{b} + \bar{c}|^2 = |\bar{a}|^2 + |\bar{b}|^2 + |\bar{c}|^2 + 2(\bar{a}.\bar{b} + \bar{b}.\bar{c} + \bar{c}.\bar{a})$

- Let l_1, m_1, n_1 be the direction cosines of $\bar{a}$ and let l_2, m_2, n_2 be the direction cosines of $\bar{b}$ and let $(\bar{a}, \bar{b}) = \theta$ then $Cos\theta = l_1 l_2 + m_1 m_2 + n_1 n_2$
- The vector equation to the plane which is at a distance of p units from the origin and $\hat{n}$ is a unit vector perpendicular to the plane is $\bar{r}.\hat{n} = p$
- If the origin lies on the plane then its equation is $\bar{r}.\bar{n} = 0$
- The vector equation of a plane passing through the point $A(\bar{a})$ and perpendicular to the vector $\bar{n}$ is $(\bar{r} - \bar{a}).\bar{n} = \bar{0}$

W.E-6 : The vector equation of the plane passing through the point $(3,-2,1)$ and perpendicular to the vector $(4,7,-4)$ is

Sol. $\{\bar{r} - (3\bar{i} - 2\bar{j} + \bar{k})\}.(4\bar{i} + 7\bar{j} - 4\bar{k}) = 0$

$\Rightarrow \bar{r}.(4\bar{i} + 7\bar{j} - 4\bar{k}) =$

$(3\bar{i} - 2\bar{j} + \bar{k}).(4\bar{i} + 7\bar{j} - 4\bar{k}) = 12 - 14 - 4$

$\Rightarrow \bar{r}.(4\bar{i} + 7\bar{j} - 4\bar{k}) = -6$

- In a parallelogram, if its diagonals are equal then it is a rectangle.
- In a parallelogram, the sum of the squares of the diagonals is equal to the sum of the squares of the sides.
- If $\bar{F}$ be the force and $\bar{s}$ be the displacement inclined at an angle θ with the direction of the force, then work done $\bar{F}.\bar{S}$
- If a constant force $\bar{F}$ acting on a particle displaces it from A to B, then work done, $W = \bar{F}.\overline{AB}$
- If $\bar{F}$ is the resultant of the forces $\bar{F}_1, \bar{F}_2 ......\bar{F}_n$ then work done in displacing the particle from A to B is $W = (\bar{F}_1 + \bar{F}_2 + ... + \bar{F}_n).\overline{AB}$
- If a number of forces are acting on a particle, the sum of the work done by the seperate forces is equal to the work done by the resultant force.
- A line makes angles $\alpha, \beta, \gamma, \delta$, with the four diagonals of a cube then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma + \cos^2\delta = 4/3$
- If $\bar{r}$ is any vector then $\bar{r} = (\bar{r}.\bar{i})\bar{i} + (\bar{r}.\bar{j})\bar{j} + (\bar{r}.\bar{k})\bar{k}$.
- If $\bar{a}, \bar{b}$ are two vectors then

  i) $\bar{a}.\bar{a} \ge 0$ ii) $|\bar{a}.\bar{b}| \le |\bar{a}||\bar{b}|$

  iii) $|\bar{a} + \bar{b}| \le |\bar{a}| + |\bar{b}|$ iv) $|\bar{a} - \bar{b}| \ge ||\bar{a}| - |\bar{b}||$
- The cartesian equation of the plane passing through the point $A(x_1, y_1, z_1)$ and perpendicular to the vector $\bar{m} = a\bar{i} + b\bar{j} + c\bar{k}$ is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$.
- The equation of the plane passing through the point $A(x_1, y_1, z_1)$ and whose normal has d.r.s a,b,c is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$.
- Angle between any two diagonals of a cube is $\cos^{-1}(1/3)$.
- Angle between a diagonal of a cube and a diagonal of a face of the cube which are passes through the same corner is $\cos^{-1}\sqrt{2/3}$.
- Angle between a diagonal of a cube and edge of a cube is $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Angle between a line and a plane :

- i) The angle between a line and a plane is the complement of the angle between the line and normal to the plane. If θ is the anlge between a line $\bar{r} = \bar{a} + t\bar{b}$ and a plane $\bar{r}.\bar{m} = d$ then

$$\cos(90^0 - \theta) = \sin\theta = \frac{\bar{b}.\bar{m}}{|\bar{b}||\bar{m}|}.$$

WE-7: The angle between the line $\bar{r} = (-i + 3j + 3k) + t(2i + 3j + 6k)$ and the plane $\bar{r}.(-\bar{i} + \bar{j} + \bar{k}) = 5$ is

Sol: $\sin\theta = \dfrac{(2\bar{i}+3\bar{j}+6\bar{k}).(-\bar{i}+\bar{j}+\bar{k})}{|2\bar{i}+3\bar{j}+6\bar{k}||-\bar{i}+\bar{j}+\bar{k}|} =$

$$\frac{-2+3+6}{\sqrt{4+9+36}\sqrt{1+1+1}} = \frac{1}{\sqrt{3}} \Rightarrow \theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

➢ If θ is the angle between the line

$\dfrac{x-x_1}{l} = \dfrac{y-y_1}{m} = \dfrac{z-z_1}{n}$ and the plane

$ax+by+cz+d=0$

then i) $\sin\theta = \dfrac{al+bm+cn}{\sqrt{a^2+b^2+c^2}\sqrt{l^2+m^2+n^2}}$

ii) If the line is perpendicular to the plane then

$\dfrac{l}{a} = \dfrac{m}{b} = \dfrac{n}{c}$.

iii) If the line is parallel to the plane then

$al+bm+cn=0$

➢ The perpendicular distance of the plane $\bar{r}.\bar{n} = \bar{a}.\bar{n}$.

from the origin is $\dfrac{|\bar{a}.\bar{n}|}{|\bar{n}|}$

➢ Angle between the planes

$\bar{r}.\bar{n}_1 = p_1, \bar{r}.\bar{n}_2 = p_2,$ *then* $Cos\theta = \dfrac{\bar{n}_1.\bar{n}_2}{|\bar{n}_1||\bar{n}_2|}$

WE-8: Angle between the planes

$\bar{r}.(2\bar{i}-\bar{j}+\bar{k}) = 3$ ***and*** $\bar{r}.(\bar{i}+\bar{j}+2\bar{k}) = 4$ ***is***

Sol: Let $\bar{a} = 2\bar{i} - \bar{j} + \bar{k}$ and

$\bar{b} = \bar{i} + \bar{j} + 2\bar{k}$

If θ is the angle between the planes then

$$\cos\theta = \cos(\bar{a},\bar{b}) = \frac{\bar{a}.\bar{b}}{|\bar{a}||\bar{b}|} =$$

$$\frac{2-1+2}{\sqrt{4+1+1}\sqrt{1+1+4}} = \frac{3}{6} = \frac{1}{2} \Rightarrow \theta = 60°$$

C.U.Q

1. **If $|\bar{a}+\bar{b}| < |\bar{a}-\bar{b}|$ then the angle between the vectors a,b is**

1) Acute Angle 2) Obtuse Angle
3) Right Angle 4) 45^0

2. $R(\bar{r})$ **is any point on the semi-circle** $P(\bar{p})$ **and** $Q(\bar{q})$ **are the position vectors of diameter of that semicircle. Then** $\overline{PR}.\overline{QR}$ **is equal to**

1) 1 2) 0 3) 3 4) Not defined

3. **The non-zero vectors** $\bar{a}, \bar{b}, \bar{c}$ **are related by** $\bar{a} = 8\bar{b}$ **and** $\bar{c} = -7\bar{b}$ **, then the angle between** $\bar{a}$ & $\bar{c}$ **is** **(AIE-2008)**

1) 0 2) $\dfrac{\pi}{4}$ 3) $\dfrac{\pi}{2}$ 4) π

4. **If** $(\bar{a},\bar{b}) = 0^0$ *or* 180^0 *then* $\bar{a},\bar{b}$ *are*

1) Perpendicular 2) Parallel
3) Parallel and are in the same direction
4) Parallel and are in the opposite direction

5. **If** $\bar{a}.\bar{b} = -|\bar{a}||\bar{b}|$ **then the vectors** $\bar{a}$ *and* $\bar{b}$ **are**

1) Like vectors 2) Unlike vectors
3) Equal vectors 4) Perpendicular vectors

6. **If** $\bar{a},\bar{b},\bar{c}$ **are three non-zero vectors then** $\bar{a},\bar{b}$ **=** $\bar{a},\bar{c}$ **implies that**

1) $\bar{a}$ is orthogonal to $(\bar{b}+\bar{c})$
2) $\bar{a}$ is orthogonal to both $\bar{b}$ and $\bar{c}$
3) $\bar{b}=\bar{c}+\bar{a}$
4) $\bar{a}$ is orthogonal to $(\bar{b}-\bar{c})$ (or) $\bar{b} = \bar{c}$

7. **Equality holds in the triangle inequality** $|\bar{a}+\bar{b}| \le |\bar{a}|+|\bar{b}|$ *if*

1) $\bar{a} = m\bar{b}$ 2) $\bar{a} = m\bar{b}$, m>0
3) $\bar{a} = m\bar{b}$, m<0 4) $(\bar{a},\bar{b}) = 90^0$

8. **The vector equation** $\bar{a}.\bar{b} = \bar{a}.\bar{c}$ **need not always imply**

1) $\bar{a} = 0$ 2) $\bar{b} = \bar{c}$

3) $\bar{a} \perp (\bar{b}-\bar{c})$ 4) All the above

9. **Let $\bar{a}$ and $\bar{b}$ be two nonzero vectors.Then $(\bar{a}.\bar{b})^2$ is**

1) $\geq |\bar{a}|^2 |\bar{b}|^2$ 2) $= |\bar{a}|^2 |\bar{b}|^2$ 3) $\leq |\bar{a}|^2 |\bar{b}|^2$ 4) $= |\bar{a}|^2 |\bar{b}|$

10. **If $\bar{a},\bar{b},\bar{c}$ are mutually perpendicular and $\bar{r}$ is any vector, then $\Sigma\left(\frac{\bar{r}.\bar{a}}{\bar{a}.\bar{a}}\right)\bar{a} =$**

1) $\bar{r}$ 2) $\bar{a}$ 3) $\bar{b}$ 4) $\bar{c}$

11. **The equation of the plane passing through the point with position vector $\bar{a}$ and perpendicular to $\bar{b}$ is (EAM-1990)**

1) $\bar{r}.(\bar{a}\times\bar{b})=0$ 2) $\bar{r}=\bar{a}\times\bar{b}$

3) $\bar{r}=\bar{b}\times\bar{a}$ 4) $(\bar{r}-\bar{a}).\bar{b}=0$

12. **The angle between the straight lines $\bar{r}=\bar{a}+t\bar{u},\bar{r}=\bar{a}+s\bar{v}$ is θ then**

1) $\sin\theta=\frac{\bar{u}.\bar{v}}{|\bar{u}||\bar{v}|}$ 2) $\cos\theta=\frac{\bar{u}.\bar{v}}{|\bar{u}||\bar{v}|}$

3) $\cos\theta=\bar{u}.\bar{v}$ 4) $\sin\theta=\bar{u}.\bar{v}$

13. **If $|\bar{a}|+|\bar{b}|=|\bar{c}|$ and $\bar{a}+\bar{b}=\bar{c}$ then the angle between $\bar{a}$ and $\bar{b}$**

1) $0°$ 2) $\frac{\pi}{6}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$

C.U.Q-KEY

01) 2	02) 2	03) 4	04) 2
05) 2	06) 4	07) 2	08) 4
09) 3	10) 1	11) 4	12) 2
13) 1			

C.U.Q-HINTS

1. squaring and expand
2. Angle in a semicircle is a right angle.
3. since $\bar{a}$ & $\bar{c}$ have opposite direction $(\bar{a},\bar{c})=\pi$.
4. Definition of parallel vectors
5. $\cos\theta=-1 \Rightarrow \theta=180°$
6. $\bar{a}.\bar{b}-\bar{a}.\bar{c}=0 \Rightarrow \bar{a}.(\bar{b}-\bar{c})=0$
7. $(a,b)=0 \Rightarrow$ vector inequality concept $a=mb, m>0$
8. $\bar{a}.\bar{b}=\bar{a}.\bar{c}$ then (1) $\bar{a}=0$ (2) $\bar{b}=\bar{c}$ (3) $\bar{a}\perp\bar{b},\bar{c}$ the three results may be true or may not be true.
9. $|\bar{a}.\bar{b}| \leq |a||b| \Rightarrow (a.b)^2 \leq |a|^2|b|^2$
10. $\bar{r}=\Sigma(\bar{r}.\hat{a})\hat{a}$
11. Vector equation of plane passing through $\bar{a}$ and perpendicular $\bar{b}$ is $(\bar{r}-\bar{a}).\bar{b}=0$ or $\bar{r}.\bar{b}=\bar{a}.\bar{b}$
12. $\cos\theta=\frac{\bar{u}.\bar{v}}{|\bar{u}||\bar{v}|}$ of the lines passing through $\bar{a}$ and parallel to $\bar{u},\bar{v}$.
13. $\bar{a}+\bar{b}=\bar{c}$ Squaring on both sides

LEVEL - I (C.W)

Scalar (or) Dot Product of Vectors Geometrical Interpretation Orthogonal Projection

1. **If $\bar{a}=\bar{i}+2\bar{j}+\bar{k}$,$\bar{b}=2\bar{j}+\bar{k}-\bar{i}$, then component of $\bar{a}$ perpendicular to $\bar{b}$ is**

1) $\frac{5}{2}\bar{i}+\frac{2}{3}\bar{j}+\frac{1}{2}\bar{k}$ 2) $\frac{5}{3}\bar{i}+2\bar{j}+\frac{1}{3}\bar{k}$

3) $\frac{5\bar{i}+2\bar{j}+\bar{k}}{3}$ 4) $\bar{i}+\bar{j}+\bar{k}$

2. **The orthogonal projection of $\bar{a}=2\bar{i}+3\bar{j}+3\bar{k}$ on $\bar{b}=\bar{i}-2\bar{j}+\bar{k}$ (where $\bar{i}.\bar{j}.\bar{k}$ are unit vectors along three mutually perpendicular directions) is (EAM-1996)**

1) $\frac{-\bar{i}+2\bar{j}-\bar{k}}{6}$ 2) $\frac{-\bar{i}+2\bar{j}-\bar{k}}{\sqrt{6}}$

3) $\bar{i}-2\bar{j}+\bar{k}$ 4) $-\bar{i}+2\bar{j}-\bar{k}$

3. **Given two vectors $\bar{a}=2\bar{i}-3\bar{j}+6\bar{k}$, $\bar{b}=-2\bar{i}+2\bar{j}-\bar{k}$ and**

$\lambda = \frac{\text{the projection of } \bar{a} \text{ on } \bar{b}}{\text{the projection of } \bar{b} \text{ on } \bar{a}}$, **then the value of λ is**

1) $\frac{3}{7}$ 2) 7 3) 3 4) $\frac{7}{3}$

4. If the vector OP in XY plane whose magnitude is $\sqrt{3}$ makes an angle 60^0 with Y-axis, the length of the component of the vector in direction of X-axis is

1) 1 2) $\sqrt{3}$ 3) 1/2 4) 3/2

5. If $\bar{a}$ = $4\bar{i}+6\bar{j}$ and $\bar{b}$ = $3\bar{j}+4\bar{k}$, then the vector form of the component of a along b is

1) $\frac{18(3\bar{i}+4\bar{k})}{10\sqrt{3}}$ 2) $\frac{18(3\bar{j}+4\bar{k})}{25}$

3) $\frac{18(3\bar{i}+4\bar{k})}{\sqrt{13}}$ 4) $\frac{(3\bar{j}+4\bar{k})}{25}$

6. If $\bar{p}=(2,1,3), \bar{q}=(-2,3,1), \bar{r}=(3,-2,4)$ and $\bar{j}$ is the unit vector in the direction of $y-$axis, then $(2\bar{p}+3\bar{q}-4\bar{r}).\bar{j}=$

1) 18 2) 19 3) 20 4) 21

7. The projection of the vector $\bar{a}=4\bar{i}-3\bar{j}+2\bar{k}$ on the vector making equal angles (acute) with coordinate axes and having magnitude $\sqrt{3}$ is

1) 3 2) $\sqrt{3}$ 3) $2\sqrt{3}$ 4) 1

8. The sum of the length of projections of $p\bar{i}+q\bar{j}+r\bar{k}$ on the coordinate axes, where $p=2, q=3$ and $r=1$ is

1) 6 2) 5 3) 4 4) 3

9. Let P = (1,0,-1) Q = (-1,2,0) R = (2,0,-3) S=(3,-2,-1), then the length of the component of RS on PQ is

1) 1/3 2) 2/3 3) 4/3 4) 5/3

10. The vectors $\bar{a},\bar{b},\bar{c}$ are of the same length and taken pair wise, they form equal angles. If $\bar{a}=\bar{i}+\bar{j}$ and $\bar{b}=\bar{j}+\bar{k}$ then the components of $\bar{c}$ are

1) (1, 0, 1) 2) (1, 2, 3)

3) (–1, 1, 2) 4) (–1, 4, 1)

Properties of dot product:

11. If $\bar{a}$ is collinear with $\bar{b}=3\bar{i}+6\bar{j}+6\bar{k}$ and $\bar{a}.\bar{b}=27$. Then $\bar{a}$ is equal to

1) $3(\bar{i}+\bar{j}+\bar{k})$ 2) $\bar{i}+3\bar{j}+3\bar{k}$

3) $\bar{i}+2\bar{j}+2\bar{k}$ 4) $2\bar{i}+2\bar{j}+2\bar{k}$

12. Let a and b be two unit vectors and θ be the angle between them then $|\bar{a}+\bar{b}|^2 - |\bar{a}-\bar{b}|^2 =$

1) $\cos\theta$ 2) $2\cos\theta$ 3) $3\cos\theta$ 4) $4\cos\theta$

13. The vector $\bar{x}$ which is perpendicular to (2,-3,1) and (1,-2,3) and which satisfies the condition $\bar{x}.(\bar{i}+2\bar{j}-7k)=10$

1) $3\bar{i}+5\bar{j}+\bar{k}$ 2) $7\bar{i}-5\bar{j}+\bar{k}$

3) $3\bar{i}-5\bar{j}+\bar{k}$ 4) $7\bar{i}+5\bar{j}+\bar{k}$

14. $A = 4\bar{i}+5\bar{j}+6\bar{k}$, $B = 5\bar{i}+6\bar{j}+4\bar{k}$

$C=6\bar{i}+4\bar{j}+5\bar{k}$ **are the vertices of**

1) Scalen triangle 2) Equilateral triangle

3) Right angled triangle 4) Isosceles triangle

15. If θ is acute angle and the vector $(\sin\theta)\bar{i}+(\cos\theta)\bar{j}$ is perpendicular to the vector $\bar{i}-\sqrt{3}\bar{j}$ then $\theta=$ (EAM-2000)

1) $\frac{\pi}{6}$ 2) $\frac{\pi}{5}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{3}$

16. $\bar{a},\bar{b},\bar{c}$ are three vectors, such that $\bar{a}+\bar{b}+\bar{c}=\bar{0}, |\bar{a}|=1, |\bar{b}|=2, |\bar{c}|=3$, then $(\bar{a}.\bar{b}+\bar{b}.\bar{c}+\bar{c}.\bar{a})^2$ is equal to

1) – 7 2) 49 3) 7 4) 1

17. If $\bar{A}.(\bar{B}+\bar{C}) = \bar{B}.(\bar{C}+\bar{A}) = \bar{C}.(\bar{A}+\bar{B}) = 0$ and $|\bar{A}|=3, |\bar{B}|=4$ and $|\bar{C}|=5$ then $|\bar{A}+\bar{B}+\bar{C}|=$

1) 5 2) $5\sqrt{2}$ 3) $5/\sqrt{2}$ 4) $\sqrt{2}$

18. If $|\mathbf{a}|=3, |\mathbf{b}|=4$ and $|\mathbf{a}-\mathbf{b}|=5$ then $|\mathbf{a}+\mathbf{b}|=$ (EAM 1994)

1) 6 2) 5 3) 4 4) 3

19. If $|\bar{a}|=1, |\bar{b}|=2, |\bar{a}-\bar{b}|^2+|\bar{a}+2\bar{b}|^2=20$, then $(\bar{a},\bar{b})=$

1) $\frac{\pi}{3}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{6}$ 4) $\frac{2\pi}{3}$

20. Let $|\bar{a}|=3$ and $|\bar{b}|=4$. The value of 'μ' for which the vectors $\bar{a}+\mu\bar{b}$ and $\bar{a}-\mu\bar{b}$ are perpendicular is...

1) $\frac{3}{4}$ 2) $\frac{2}{3}$ 3) $\pm\frac{3}{4}$ 4) $-\frac{2}{3}$

21. If $\bar{a},\bar{b},\bar{c},\bar{d}$ are the vertices of a square then

1) $\bar{b}-\bar{a}=\bar{c}-\bar{b}$ 2) $\bar{a}+\bar{b}=\bar{c}$
3) $(\bar{c}-\bar{a}).(\bar{d}-\bar{b})=0$ 4) $\bar{a}+\bar{b}=0$

22. If $\bar{a}+\bar{b}$ is perpendicular to $\bar{b}$ and $\bar{a}+2\bar{b}$ is perpendicular to $\bar{a}$ then.

1) $|\bar{a}|=|\bar{b}|$ 2) $|\bar{a}|=\sqrt{2}|\bar{b}|$
3) $|\bar{b}|=\sqrt{2}|\bar{a}|$ 4) $|\bar{a}|=|\bar{b}|\sqrt{3}$

23. $(\bar{a}-\bar{d}).(\bar{b}-\bar{c})+(\bar{b}-\bar{d}).(\bar{c}-\bar{a})+(\bar{c}-\bar{d}).(\bar{a}-\bar{b})=$

1) $\bar{0}$ (null vector) 2) 0
3) $\bar{a}.\bar{b}+\bar{c}.\bar{d}$ 4) $\bar{a}.\bar{c}+\bar{b}.\bar{d}$

24. If $\bar{a},\bar{b},\bar{c}$ are unit vectors, then $|\bar{a}-\bar{b}|^2+|\bar{b}-\bar{c}|^2+|\bar{c}-\bar{a}|^2$ does not exceed

1) 4 2) 9 3) 8 4) 6

25. Dot product of a vector with vectors $3\bar{i}-5\bar{k}, 2\bar{i}+7\bar{j}$ and $\bar{i}+\bar{j}+\bar{k}$ are respectively $-1, 6$ and 5. The vector is

1) $3\bar{i}+\bar{j}+2\bar{k}$ 2) $3\bar{i}+2\bar{k}$
3) $3\bar{i}-2\bar{k}$ 4) $\bar{i}+\bar{j}+\bar{k}$

26. The vector $\bar{b}$ which is collinear with the vector $\bar{a}=(2,1,-1)$ and satisfies the relation $\bar{a}.\bar{b}=3$ is ...

1) $\left(1,\frac{1}{2},\frac{-1}{2}\right)$ 2) $\left(\frac{2}{3},\frac{1}{3},\frac{-1}{3}\right)$
3) $\left(\frac{1}{2},\frac{1}{4},\frac{-1}{4}\right)$ 4) $(1,1,0)$

27. If two adjacent sides of a square are represented by the vectors $x\bar{i}+\bar{j}+4\bar{k}$ and $3\bar{i}+y\bar{j}$ then $xy=$

1) 1 2) 2 3) 3 4) –3

28. ABCD is a rhombus. If $\overline{AC}=\bar{i}+(1+\lambda)\bar{j}+(\lambda-2)\bar{k}$ and $\overline{BD}=(2\lambda-1)\bar{i}+\bar{j}-\bar{k}$, then $\lambda=$

1) 1 2) –1 3) 2 4) –2

29. If the sum of two unit vectors is a unit vector, then the magnitude of their difference is

1) 3 2) $\sqrt{3}$ 3) $\sqrt{13}$ 4) $\sqrt{7}$

Angle between two vectors:

30. If $|\bar{a}|=|\bar{b}|=|\bar{a}-\bar{b}|$, then the angle between $\bar{a}$ and $\bar{b}$ is

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) $\frac{2\pi}{3}$

31. If $\bar{a}$ and $\bar{b}$ are two unit vectors and θ be the angle between them, then $\sin\left(\frac{\theta}{2}\right)=$

1) $|\bar{a}-\bar{b}|$ 2) $|\bar{a}+\bar{b}|$ 3) $\frac{1}{2}|\bar{a}-\bar{b}|$ 4) $\frac{1}{2}|\bar{a}+\bar{b}|$

32. If a and b are unit vectors and θ is the angle between them ,then a-b will be a unit vector if $\theta=$ (EAM-1997)

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{6}$ 4) $\frac{\pi}{2}$

33. If $\bar{a}=p\bar{i}+3\bar{j}-7\bar{k}, \bar{b}=p\bar{i}-p\bar{j}+4\bar{k}$ and if the angle between a and b is acute , then the values of p lies in

1) $P<-4$ or $p>7$ 2) (-7,4)
3) $P\le -4 or p\ge 7$ 4) [-7,4]

34. If $\bar{a}+\bar{b}+\bar{c}=0, |\bar{a}|=3, |\bar{b}|=5$ and $|\bar{c}|=7$ then the angle between $\bar{a}$ and $\bar{b}$ is

1) 30° 2) 45° 3) 60° 4) 90°

35. If $\bar{a},\bar{b},\bar{c}$ are three unit vectors such that $|\bar{a}+\bar{b}+\bar{c}|=1$ and $\bar{a}\perp\bar{b}$. If $\bar{c}$ makes angles α,β with $\bar{a},\bar{b}$ respectively, then $\cos\alpha+\cos\beta$ is equal to

1) $\frac{3}{2}$ 2) 1 3) -1 4) $\frac{-3}{2}$

36. If $\overline{AB}$ =(3,-2,2), $\overline{BC}$=(-1,0-2) are the adjacent sides of a parallelogram, then the obtuse angle between its diagonals is

1) $\frac{\pi}{4}$ 2) $\pi/2$ 3) $\pi/3$ 4) $\frac{3\pi}{4}$

37. If $\bar{a},\bar{b},\bar{c}$ are three mutually perpendicular vectors such that $|\bar{a}|=|\bar{b}|=|\bar{c}|$ then $(\bar{a}+\bar{b}+\bar{c},\bar{a})=$

1) $\frac{\pi}{3}$ 2) $\cos^{-1}\left(\frac{1}{3}\right)$

3) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$ 4) $\cos^{-1}\left(\frac{2}{3}\right)$

38. If $\bar{a}$ and $\bar{b}$ are unit vectors inclined to x-axis at angle 30^0 and 120^0, then $|\bar{a}+\bar{b}|$ equals

1) $\sqrt{2/3}$ 2) $\sqrt{2}$ 3) $\sqrt{3}$ 4) 2

39. If $\overline{AB}=-\bar{i}-2\bar{j}-6\bar{k}$, $\overline{BC}=2\bar{i}-\bar{j}+\bar{k}$, $\overline{AC}=\bar{i}-3\bar{j}-5\bar{k}$ .Then $\angle B=$

1) $Cos^{-1}\left(\sqrt{\frac{40}{41}}\right)$ 2) $Cos^{-1}\left(\sqrt{\frac{6}{41}}\right)$

3) $Cos^{-1}\left(\frac{6}{41}\right)$ 4) $Cos^{-1}\left(\frac{62}{63}\right)$

40. If $\bar{v}$ and $\bar{w}$ are two mutually perpendicular unit vectors and $\bar{u}=a\bar{v}+b\bar{w}$, where a and b are non zero real numbers, then the angle between $\bar{u}$ and $\bar{w}$ is

1) $\cos^{-1}\left(\frac{b}{\sqrt{a^2+b^2}}\right)$ 2) $\cos^{-1}\left(\frac{a}{\sqrt{a^2+b^2}}\right)$

3) $\cos^{-1}(b)$ 4) $\cos^{-1}(a)$

41. The value of a, for which the points A,B, C with position vectors $2\bar{i}-\bar{j}+\bar{k},\bar{i}-3\bar{j}-5\bar{k}$ and $a\bar{i}-3\bar{j}+\bar{k}$ respectively are the vertices of a right angled triangle with $\angle C=\frac{\pi}{2}$ are **(AIE-2006)**

1) -2 or 1 2) 2 or -1
3) 2 or 1 4) -2 or -1

42. The points O, A, B, C, D are such that $\overline{OA}=\bar{a},\overline{OB}=\bar{b},\overline{OC}=2\bar{a}+3\bar{b}$ and $\overline{OD}=\bar{a}-2\bar{b}$. If $|\bar{a}|=3|\bar{b}|$, then the angle between $\overline{BD}$ and $\overline{AC}$ is

1) π 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{6}$

43. If $\bar{a}$ =(-1,1,2); $\bar{b}$ =(2,1,-1); $\bar{c}$ =(-2,1,3) then the angle between $2\bar{a}-\bar{c}$ and $\bar{a}+\bar{b}$ is

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{6}$ 4) $\frac{3\pi}{2}$

Vector Equation of a Plane Normal form

44. The cartesian equation of the plane perpendicular to vector $3\bar{i}-2\bar{j}-2\bar{k}$ and passing through the point $2\bar{i}+3\bar{j}-\bar{k}$ is

1) 3x+2y+2z=2 2) 3x-2y+2z=2
3) 3x+2y-2z=2 4) 3x-2y-2z=2

45. The perpendicular distance from origin to the plane 3x - 2y - 2z = 2 is

1) $1/\sqrt{17}$ 2) $2/\sqrt{17}$ 3) $3/\sqrt{17}$ 4) $4/\sqrt{17}$

46. The vector equation of the plane which is perpendicular to 2i-3j+k and at a distance of 5 units from the origin is (EAM-1991)

1) $\bar{r}.(2\bar{i}-3\bar{j}+\bar{k})=5\sqrt{14}$ 2) $\bar{r}.(2\bar{i}-3\bar{j}+\bar{k})=5$

3) $\bar{r}.\frac{(2\bar{i}-3\bar{j}+\bar{k})}{\sqrt{14}}$ 4) $\frac{\bar{r}.(2\bar{i}+3\bar{j}+\bar{k})}{\sqrt{14}}$

Angle between the planes:

47. The angle between planes

$\bar{r}.(2\bar{i}-3\bar{j}+4\bar{k})+11=0$ **and** $\bar{r}.(3\bar{i}-2\bar{j}-3\bar{k})+27=0$ **is**

1) $\pi/6$ 2) $\pi/3$ 3) $\pi/4$ 4) $\pi/2$

Applications of Scalar Product in Mechanics to find the work done:

48. A particle is acted upon by constant forces $4\bar{i}+\bar{j}-3\bar{k}$ and $3\bar{i}+\bar{j}-\bar{k}$ which displace it from a point $\bar{i}+2\bar{j}+3\bar{k}$ to the point $5\bar{i}+4\bar{j}+\bar{k}$. The work done in standard units by the forces is given by (AIE-2004)

1) 40 2) 30 3) 25 4) 15

Locus:

49. The locus of the point equidistant from two given points a and b is given by

1) $[\bar{r}-1/2(\bar{a}+\bar{b})].(\bar{a}-\bar{b})=0$

2) $[\bar{r}-1/2(\bar{a}-\bar{b})].(\bar{a}+\bar{b})=0$

3) $[\bar{r}-1/2(\bar{a}+\bar{b})].(\bar{a}+\bar{b})=0$

4) $[\bar{r}-1/2(\bar{a}-\bar{b})].(\bar{a}-\bar{b})=0$

50. If the vectors $\bar{i}-2x\bar{j}-3y\bar{k}$ and $\bar{i}+3x\bar{j}-2y\bar{k}$ are orthogonal to each other,then the locus of the point (x,y) is

1) Circle 2) Ellipse

3) Hyperbola 4) Pair of lines

LEVEL-I (C.W)-KEY

01) 3	02) 1	03) 4	04) 4	05) 2
06) 2	07) 2	08) 1	09) 3	10) 1
11) 3	12) 4	13) 4	14) 2	15) 4
16) 2	17) 2	18) 2	19) 4	20) 3
21) 3	22) 2	23) 2	24) 2	25) 2
26) 1	27) 4	28) 2	29) 2	30) 3
31) 3	32) 2	33) 1	34) 3	35) 3
36) 4	37) 3	38) 2	39) 2	40) 1
41) 3	42) 2	43) 3	44) 4	45) 2
46) 1	47) 4	48) 1	49) 1	50) 3

LEVEL-I (C.W)-HINTS

1. $\bar{a}-\dfrac{(\bar{a}.\bar{b})\bar{b}}{|\bar{b}|^2}$

2. Orthogonal projection of $\bar{a}$ on $\bar{b}=\dfrac{(\bar{a}.\bar{b})\bar{b}}{|\bar{b}|^2}$

3. $\lambda=\dfrac{\frac{\bar{a}.\bar{b}}{|\bar{b}|}}{\frac{\bar{b}.\bar{a}}{|\bar{a}|}}=\dfrac{|\bar{a}|}{|\bar{b}|}$

4. $\sqrt{3}\cos(90^0-60^0)$

5. $\dfrac{(\bar{a}.\bar{b})\bar{b}}{|\bar{b}|^2}$

6. $2\bar{p}+3\bar{q}-4\bar{r}=-14\bar{i}+19\bar{j}-7\bar{k}$

7. $\dfrac{\frac{\sqrt{3}(\bar{i}+\bar{j}+\bar{k})}{\sqrt{3}}.\bar{a}}{|\bar{i}+\bar{j}+\bar{k}|}=\sqrt{3}$

8. Use $p+q+r$

9. $\dfrac{|\overline{RS}.\overline{PQ}|}{|\overline{PQ}|}$

10. $|\bar{a}|=\sqrt{2},|\bar{b}|=\sqrt{2}$, $|(1,0,1)|=\sqrt{1+0+1}=\sqrt{2}$

11. $\bar{a}=x\bar{i}+y\bar{j}+z\bar{k}$ and $\bar{a}.\bar{b}=27$.

12. $|\bar{a}+\bar{b}|^2=|\bar{a}|^2+|\bar{b}|^2+2\bar{a}.\bar{b}$

13. $\bar{x}=x\bar{i}+y\bar{j}+z\bar{k}$ and $\bar{x}.(\bar{i}+2\bar{j}-7\bar{k})=10$

14. find $|\bar{a}|,|\bar{b}|,|\bar{c}|$

15. $(\sin\theta)(1)+\cos\theta(-\sqrt{3})=0$

$\Rightarrow\dfrac{\sin\theta}{\cos\theta}=\sqrt{3}\Rightarrow\theta=60^o$

16. $|\bar{a}+\bar{b}+\bar{c}|=0\Rightarrow 1+4+9+2(\bar{a}\bar{b}+\bar{b}\bar{c}+\bar{c}\bar{c})=0$

$\Rightarrow\bar{a}.\bar{b}+\bar{b}.\bar{c}+\bar{c}.\bar{a}=-7$

17. $2(\bar{A}.\bar{B}+\bar{B}.\bar{C}+\bar{C}.\bar{A})=0$

$|\bar{A}+\bar{B}+\bar{C}|^2=9+16+25$

18. $\left|\bar{a}+\bar{b}\right|^2+\left|\bar{a}-\bar{b}\right|^2=2\left(\left|\bar{a}\right|^2+\left|\bar{b}\right|^2\right)$

19. On expanding $\bar{a}.\bar{b}=-1,\cos\left(\bar{a},\bar{b}\right)=-\frac{1}{2}$

$\left(\bar{a},\bar{b}\right)=\frac{2\pi}{3}$

20. Use $\left|\bar{a}\right|^2=\left|\mu^2\right|\left|\bar{b}\right|^2$

21. Diagonals are Perpendicular $\overline{AC}.\overline{BD}=0$

22. $\left(\bar{a}+\bar{b}\right).\bar{b}=0$ *and* $\left(\bar{a}+2\bar{b}\right).\bar{a}=0$ *simplify*

23. $\bar{a}.\bar{b}-\bar{a}.\bar{c}-\bar{d}.\bar{b}+\bar{d}.\bar{c}+\bar{b}.\bar{c}-$
$\bar{b}.\bar{a}-\bar{d}.\bar{c}+\bar{d}.\bar{a}+\ \bar{c}.\bar{a}-\bar{c}.\bar{b}-\bar{d}.\bar{a}+\bar{d}.\bar{b}=0$

24. $\left|\bar{a}+\bar{b}+\bar{c}\right|^2\geq 0\Rightarrow\bar{a}.\bar{b}+\ \bar{b}.\bar{c}+\bar{c}.\bar{a}\geq-\frac{3}{2}$

Now, $\left|\bar{a}-\bar{b}\right|^2+\left|\bar{b}-\bar{c}\right|^2+\left|\bar{c}-\bar{a}\right|^2\leq 9$

25. Check the dot product from options.

26. Check from options that $\bar{b}=\lambda\bar{a}$ and $\bar{a}.\bar{b}=3$

27. dot product = 0

28. $\overline{AC}.\overline{BD}=0$

29. *use* $|\bar{a}+\bar{b}|^2+|\bar{a}-\bar{b}|^2\ =2\left(|\bar{a}|^2+|\bar{b}|^2\right)$

30. $|\bar{a}-\bar{b}|^2=|\bar{a}|^2+|\bar{b}|^2-2|\bar{a}||\bar{b}|\cos\theta$

$\Rightarrow|\bar{a}|^2=|\bar{a}|^2+|\bar{b}|^2-2|\bar{a}||\bar{b}|\cos\theta$

$+|\bar{b}|^2=+2|\bar{a}||\bar{b}|\cos\theta$, $\cos\theta=\frac{1}{2}\Rightarrow\theta=60^0$

31. $\sin\frac{\theta}{2}=\frac{1}{2}\left|\bar{a}-\bar{b}\right|$

32. $\left|\bar{a}-\bar{b}\right|=1\Rightarrow\left|\bar{a}-\bar{b}\right|^2=1$

33. $\bar{a}.\bar{b}>0$

34. $\bar{a}+\bar{b}+\bar{c}=\bar{0}\qquad\Rightarrow\bar{a}+\bar{b}=-\bar{c}$
S.O.B.S and expand

35. $\left|\bar{a}+\bar{b}+\bar{c}\right|^2=\left|\bar{a}\right|^2+\left|\bar{b}\right|^2+\left|\bar{c}\right|^2+2\left(\bar{a}.\bar{b}+\bar{b}.\bar{c}+\bar{c}.\bar{a}\right)$

36. Find $\overline{AC}\ \&\ \overline{BD}$ $\therefore\cos\theta=\frac{\overline{AC}.\overline{BD}}{\left|\overline{AC}\right|\left|\overline{BD}\right|}$

37. $\cos\left(\bar{a}+\bar{b}+\bar{c},\bar{a}\right)=\frac{\left(\bar{a}+\bar{b}+\bar{c}\right).\bar{a}}{\left|\bar{a}+\bar{b}+\bar{c}\right|\left|\bar{a}\right|}$

$=\frac{\left|\bar{a}\right|^2}{\sqrt{3}\left|\bar{a}\right|^2}=\frac{1}{\sqrt{3}}$

38. Clearly, $\bar{a}$ and $\bar{b}$ are at right angles

$\therefore\left|\bar{a}+\bar{b}\right|^2=\left|\bar{a}\right|^2+\left|\bar{b}\right|^2+2\left|\bar{a}\right|\left|\bar{b}\right|\cos 90°$

$\Rightarrow\left|\bar{a}+\bar{b}\right|^2=1+1+0=2\Rightarrow\left|\bar{a}+\bar{b}\right|=\sqrt{2}$

39. $\cos\theta=\frac{\overline{BC}.\overline{BA}}{\left|\overline{BC}\right|\left|\overline{BA}\right|}$

40. $\bar{u}\ .\ \bar{\omega}=a\left(\bar{v}.\bar{\omega}\right)+b\left(\bar{\omega}.\bar{\omega}\right)=b$

$\Rightarrow\cos\theta=\frac{b}{|\bar{u}||\bar{\omega}|}=\frac{b}{\sqrt{a^2+b^2}}$

41. $\overline{CA}.\overline{CB}=0$

42. Use $\cos\theta=\frac{\overline{BD}.\overline{AC}}{\left|\overline{BD}\right|\left|\overline{AC}\right|}$

43. $2\bar{a}-\bar{c}=\left(0,1,1\right)$, $\bar{a}+\bar{b}=\left(1,2,1\right)$

then $\cos\theta=\frac{2+1}{\sqrt{2}\sqrt{6}}\Rightarrow\cos^{-1}\left(\sqrt{3}/2\right)$, $\theta=30^0$

44. Use $\left(\bar{r}-\bar{a}\right).\bar{b}=0$

45. Distance = $\frac{|d|}{\sqrt{a^2+b^2+c^2}}$

46. $\bar{a}=2i-3\bar{j}+\bar{k};\ p=5\Rightarrow$ plane equation

$\bar{r}.\frac{\bar{a}}{\left|\bar{a}\right|}=p$

47. $\frac{\bar{m}_1.\bar{m}_2}{\left|\bar{m}_1\right|\left|\bar{m}_2\right|}=\cos\theta$

48. $\bar{F}=\bar{F}_1+\bar{F}_2$ and W.E. = $\bar{F}\ .\ \bar{d}$

49. The locus of a point which is equidistant from the given two points is the perpendicular bisector of the line segment joining the two points i.e., $\overline{MP}.\overline{BA}=0$ ($\because M$ is mid point)

50. $\bar{a}.\bar{b}=0$

$6x^2-6y^2=1$, *hyperbola*

LEVEL - I (H.W)

Scalar (or) Dot Product of Vectors Geometrical Interpretation Orthogonal Projection:

1. If $\bar{a}=2\bar{i}+\bar{j}+2\bar{k}$ and $\bar{b}=5\bar{i}-3\bar{j}+\bar{k}$, then the projection of $\bar{b}/2$ on $\bar{a}$ is
   1) 1 2) 2 3) 3/2 4) 4
2. If $\bar{a}=2\bar{i}+3\bar{j}$ and $\bar{b}=3\bar{j}+4\bar{k}$. Then the vector form of the component of $\bar{a}$ along $\bar{b}$ is
   1) $\frac{9(3\bar{j}+4\bar{k})}{10\sqrt{3}}$ 2) $\frac{9(3\bar{j}+4\bar{k})}{25}$
   3) $\frac{9(3\bar{j}+4\bar{k})}{\sqrt{13}}$ 4) $3\bar{j}+4\bar{k}$
3. If $\bar{a}=2\bar{i}+\bar{j}+\bar{k}, \bar{b}=\bar{i}+5\bar{j}, \bar{c}=4\bar{i}+4\bar{j}-2\bar{k}$ then the length of the projection of $(3\bar{a}-2\bar{b})$ in the direction of $\bar{c}$
   1) 3 2) –3 3) 33 4) –33
4. The angle between $\bar{a}$ and $\bar{b}$ is $\frac{5\pi}{6}$ and the projection of $\bar{a}$ on $\bar{b}$ is $\frac{-6}{\sqrt{3}}$ then $|\bar{a}|=$
   1) 6 2) $\frac{\sqrt{3}}{2}$ 3) 12 4) 4
5. The component of $a\bar{i}+b\bar{j}+c\bar{k}$ on the Y–axis is
   1) a 2) b 3) c 4) 0
6. The components of a vector on the co-ordinate axes are 2,1,2. Then the length of the vector is
   (1) 3 (2) 1 (3) 2 (4) 5
7. A vector has 3 units length and equal components then the vector is
   (1) $\bar{i}+\bar{j}+\bar{k}$ (2) $\sqrt{2}(\bar{i}+\bar{j}+\bar{k})$
   (3) $\sqrt{3}(\bar{i}+\bar{j}+\bar{k})$ (4) $\sqrt{5}(\bar{i}+\bar{j}+\bar{k})$
8. The length of the orthogonal projection of a vector $\bar{i}-2\bar{j}+\bar{k}$ on vector $4\bar{i}-4\bar{j}+7\bar{k}$ is
   1) $\frac{19}{6}$ 2) $\frac{19}{7}$
   3) $\frac{19(4\bar{i}-4\bar{j}+7\bar{k})}{9}$ 4) $\frac{19}{9}$
9. Let $\bar{a}=\bar{i}-\bar{j}+3\bar{k}$ and $\bar{b}=3\bar{i}-5\bar{j}+6\bar{k}$. Then the magnitude of the projection of $2\bar{a}-\bar{b}$ on $\bar{a}+\bar{b}$ is
   1) $\frac{22}{\sqrt{133}}$ 2) $\frac{11\sqrt{2}}{\sqrt{10}}$ 3) $\frac{22}{\sqrt{10}}$ 4) $\frac{22}{\sqrt{5}}$
10. $\bar{a},\bar{b},\bar{c}$ are three vectors such that $|\bar{a}|=1, |\bar{b}|=2, |\bar{c}|=3,$ and $\bar{b},\bar{c}$ are prependicular to each other. If the projection of $\bar{b}$ along $\bar{a}$ is same as that of $\bar{c}$ along $\bar{a}$ then $|\bar{a}-\bar{b}+\bar{c}|$ is equal to
    1) $\sqrt{2}$ 2) $\sqrt{7}$ 3) $\sqrt{14}$ 4) 14
11. The vector component of $\bar{a}=2\bar{i}+3\bar{j}+2\bar{k}$ perpendicular to the direction of $\bar{b}=\bar{i}+2\bar{j}+\bar{k}$ is
    1) $\frac{1}{3}(\bar{i}+\bar{j}+\bar{k})$ 2) $\frac{1}{3}(\bar{i}-\bar{j}+\bar{k})$
    3) $\frac{1}{3}(\bar{i}+\bar{j}-\bar{k})$ 4) $\frac{1}{3}(-\bar{i}+\bar{j}-\bar{k})$
12. If $\bar{a}=2\bar{i}+\bar{j}+\bar{k}, \bar{b}=\bar{i}+5\bar{j},$ $\bar{c}=4\bar{i}+4\bar{j}-2\bar{k}$ then the length of the projection of $(3\bar{a}-2\bar{b})$ in the direction of $\bar{c}$
    (1) 3 (2) –3 (3) 33 (4) –33

Properties of DOT Product:

13. In $\Delta ABC, A=\bar{a}, B=\bar{b}, C=\bar{c}$. If $P(\bar{r})$ is any point in the plane of ΔABC such that $\bar{b}.\bar{c}+\bar{a}.\bar{r}=\bar{c}.\bar{a}+\bar{b}.\bar{r}=\bar{a}.\bar{b}+\bar{c}.\bar{r}$ then P is of ΔABC.
    1) In-Center 2) Circum Center

3) Ortho Center 4) Centroid

14. **If two out of the 3 vectors $\bar{a},\bar{b},\bar{c}$ are unit vectors , $\bar{a}+\bar{b}+\bar{c}=\bar{0}$ and $2(\bar{a}.\bar{b}+\bar{b}.\bar{c}+\bar{c}.\bar{a})+3=0$ then the length of the third vector is**

1) 3 2) 2 3) 1 4) 0

15. **If $\bar{a},\bar{b},\bar{c}$ are unit vectors such that $\bar{a}+\bar{b}+\bar{c}=\bar{0}$ then $\bar{a}.\bar{b}+\bar{b}.\bar{c}+\bar{c}.\bar{a}=$**

1) 3/2 2) –3/2 3) $2\left[\bar{a}\ \bar{b}\ \bar{c}\right]$ 4) 0

16. **If the scalar product of the vector $\bar{i}+\bar{j}+\bar{k}$ with the unit vector parallel to the sum of the vectors $2\bar{i}+4\bar{j}-5\bar{k}$ and $x\bar{i}+2\bar{j}+3\bar{k}$ is equal to 1 then x =**

1) 0 2) 1 3) 2 4) 3

17. **If $\bar{r}=(x+y+2)\bar{i}+(2x-y+3)\bar{j}+(x+2y+7)\bar{k}$ where $\bar{r}.\bar{i}=3$, $\bar{r}.\bar{j}=5$ then $\bar{r}.\bar{k}=$**

1) 4 2) 6 3) 9 4) 8

18. **If ABCD is a parallelogram and $\overline{AC}^2-\overline{BD}^2=K\overline{AB}$.(Orthogonal projection of $\overline{AD}$ on $\overline{AB}$) then K=**

1) 1 2) 2 3) 3 4) 4

19. **If the vectors $a\bar{i}-2\bar{j}+\bar{k},2a\bar{i}+a\bar{j}-4\bar{k}$ are adjacent sides of a rectangle, then the values of 'a' are**

1) 1,2 2) −1,2 3) 1,–2 4) 2,–2

20. **If $|\bar{a}+\bar{b}|=1,|\bar{a}-\bar{b}|=7,|\bar{a}|=4,$ then $|\bar{b}|=$**

1) 1 2) 2 3) 3 4) 5

21. **If $\bar{a}.\bar{i}=\bar{a}.(\bar{i}+\bar{j})=\bar{a}.(\bar{i}+\bar{j}+\bar{k})$ then $\bar{a}=$ (EAM-2002)**

1) $\bar{i}$ 2) $\bar{j}$ 3) $\bar{k}$ 4) $\bar{i}+\bar{j}+\bar{k}$

22. **The vector $\bar{b}$ which is collinear with the vector $\bar{a}=(2,-1,2)$ and satisfies the relation $\bar{a}.\bar{b}=18$ is**

1) $(4,-2,4)$ 2) $(2,1,-1)$

3) $(1,-1,1)$ 4) $(1,1,0)$

23. **If $\bar{a},\bar{b},\bar{c}$ are orthogonal and $\bar{r}.\bar{a}=2$ then $\bar{r}-(\bar{r}.\bar{b})\bar{b}-(\bar{r}.\bar{c})\bar{c}=$**

1) $2\bar{a}$ 2) $-2\bar{a}$ 3) $\bar{a}$ 4) $-\bar{a}$

Angle between two Vectors:

24. **If $\bar{a}$ and $\bar{b}$ are two unit vectors inclined at an angle θ to each other ,then $|\bar{a}+\bar{b}|<1$ if**

1) $\theta=\frac{\pi}{6}$ 2) $\theta=\frac{\pi}{2}$

3) $\theta=\frac{\pi}{3}$ 4) $\frac{2\pi}{3}<\theta\le\pi$

25. **Let $\bar{e}_1,\bar{e}_2$ be unit vectors which include angle θ. If $\frac{1}{2}|\bar{e}_1-\bar{e}_2|=\sin(k\theta)$, then k equal to**

1) 2 2) 1 3) $\frac{1}{2}$ 4) $\frac{1}{\sqrt{2}}$

26. **If $\bar{a}$ and $\bar{b}$ are two vectors of lengths 2 and 1 respectively and $|\bar{a}-\bar{b}|=\sqrt{3}$ then $(\bar{a},\bar{b})=$**

1) $\frac{\pi}{6}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$

27. **The angle between $8(\bar{i}-\bar{k})$ and $\bar{i}+\bar{j}+\bar{k}$ is**

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{2}$ 4) $\cos^{-1}\left(\frac{1}{\sqrt{3}}\right)$

28. **Angle between the vectors**
$\bar{a}=-\bar{i}-2\bar{j}+\bar{k}$ and $\bar{b}=x\bar{i}+\bar{j}+(x+1)\bar{k}$

1) Obtuse angle 2) Acute angle

3) Right angle 4) Depends on x

29. **If $\bar{\alpha}$ and $\bar{\beta}$ are non-zero and different vectors such that $|\bar{\alpha}+\bar{\beta}|=|\bar{\beta}-\bar{\alpha}|$ then the angle between $-\bar{\alpha}$ and $\bar{\beta}$ is**

1) $\frac{\pi}{3}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{6}$

30. **The acute angles made by the line joining**

the points $(1,-3,2)$ and $(3,-5,1)$ with the coordinate axes are

1) $\cos^{-1}\left(\frac{2}{3}\right),\cos^{-1}\left(\frac{2}{3}\right),\cos^{-1}\left(\frac{1}{3}\right)$

2) $\cos^{-1}\left(\frac{3}{2}\right),\cos^{-1}\left(\frac{2}{3}\right),\cos^{-1}\left(\frac{2}{3}\right)$

3) $\cos^{-1}\left(\frac{1}{3}\right),\cos^{-1}\left(\frac{3}{2}\right),\cos^{-1}\left(\frac{1}{3}\right)$

4) $\cos^{-1}\left(\frac{2}{3}\right),\cos^{-1}\left(\frac{3}{2}\right),\cos^{-1}\left(\frac{1}{3}\right)$

31. If the angle between the vectors $(x,3,-7)$ and $(x,-x,4)$ is obtuse, the domain of 'x' is

1) $(-4,7)$ 2) $[-4,7]$ 3) $R-[-4,7]$ 4) $R-(-4,7)$

32. If $\bar{a},\bar{b},\bar{c}$ are three vectors such that $\bar{a}=\bar{b}+\bar{c}$ and the angle between $\bar{b}$ and $\bar{c}$ is $\frac{\pi}{2}$, then (here $a=|\bar{a}|,b=|\bar{b}|,c=|\bar{c}|$)

1) $a^2=b^2+c^2$ 2) $b^2=c^2+a^2$

3) $c^2=a^2+b^2$ 4) $2a^2-b^2=c^2$

33. If the position vectors of A,B and C are respectively $2\bar{i}-\bar{j}+\bar{k},\bar{i}-3\bar{j}-5\bar{k}$ and $3\bar{i}-4\bar{j}-4\bar{k}$ then $\cos^2 A=$

1) 0 2) $\frac{6}{41}$ 3) $\frac{35}{41}$ 4) 1

34. If $\bar{a}=\bar{i}+2\bar{j}-3\bar{k}$ and $\bar{b}=3\bar{i}-\bar{j}+2\bar{k}$ then the angle between the vectors $(\bar{a}+\bar{b})$ and $(\bar{a}-\bar{b})$ is

1) $\frac{\pi}{4}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{6}$

35. If the angle between $\bar{a}$ and $\bar{b}$ is 120^0 $|\bar{a}|=3$ and $|\bar{b}|=5$, then $|\bar{a}-\bar{b}|$ is equals to

1) 6 2) 7 3) 9 4) 5

36. If $\bar{a}=2\bar{i}+2\bar{j}-\bar{k}$ and $\bar{b}=6\bar{i}-3\bar{j}+2\bar{k}$ then the angle between $\bar{a}$ and $\bar{b}$ is

1) $cos^{-1}\frac{1}{3}$ 2) $\frac{\pi}{3}$

3) $cos^{-1}\left(\frac{4}{21}\right)$ 4) $cos^{-1}\left(\frac{5\sqrt{7}}{21}\right)$

37. If $\bar{e}_1$ and $\bar{e}_2$ are unit vectors and the vectors $\bar{e}_1+2\bar{e}_2$, $5\bar{e}_1-4\bar{e}_2$ are at right angles, than the angle between $\bar{e}_1$ and $\bar{e}_2$ is

1) 30^0 2) 60^0 3) 45^0 4) 75^0

38. If $\bar{a},\bar{b},\bar{c}$ are unit vectors satisfying the relation $\bar{a}+\bar{b}+\sqrt{3}\bar{c}=0$, then the angle between $\bar{a}$ and $\bar{b}$ is

1) $\frac{\pi}{6}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{2}$

39. The angle between the vectors $\bar{a}=6\bar{i}+2\bar{j}+3\bar{k}$ and $\bar{b}=2\bar{i}-9\bar{j}+6\bar{k}$ is

1) $\cos^{-1}\left(\frac{2}{3}\right)$ 2) $\cos^{-1}\left(\frac{7}{12}\right)$

3) $\cos^{-1}\left(\frac{12}{77}\right)$ 4) $\cos^{-1}\left(\frac{1}{3}\right)$

40. Angle between $\bar{a}$ and $\bar{b}$ is 120^0. If $|\bar{b}|=2|\bar{a}|$ and the vectors $\bar{a}+x\bar{b}$, $\bar{a}-\bar{b}$ are at right angles, then x is equal to

1) $\frac{1}{3}$ 2) $\frac{1}{5}$ 3) $\frac{2}{5}$ 4) $\frac{2}{3}$

41. Let $A(2,-3,0)\,B(2,-1,1)\,C(0,1,4)$ be the vertices of ΔABC. Then angle between the median $\overline{BD}$ and the base $\overline{AC}$ is

(1) 30° (2) 45° (3) 60° (4) 90°

Vector Equation of a Plane normal form:

42. The cartesian equation of the plane passing through the point $(3,-2,1)$ and perpendicular to vector $4\bar{i}+7\bar{j}-4\bar{k}$ is

1) $4x+7y+4z-6=0$ 2) $4x+7y-4z+6=0$

3) $4x-7y-4z-6=0$ 4) $4x-7y+4z+6=0$

43. The perpendicular distance from origin to the plane $3x-4y-5z=3$ is

1) $\frac{3}{5}$ 2) 3 3) $\frac{3}{4\sqrt{2}}$ 4) $\frac{3}{5\sqrt{2}}$

44. The perpenducular distance from origin to the plane $3x+4y+5z=25$ is

(1) $\frac{3}{5}$ (2) 3 (3) $\frac{3}{4\sqrt{2}}$ (4) $\frac{5}{\sqrt{2}}$

Angle between the Planes:

45. Angle between the planes $2x-y+2z=3$, $3x+6y+2z=4$ is

1) $\cos^{-1}\left(\frac{4}{21}\right)$ 2) $\cos^{-1}\left(\frac{4}{441}\right)$

3) $\sin^{-1}\left(\frac{4}{21}\right)$ 4) $\sin^{-1}\left(\frac{4}{441}\right)$

46. Angle between the plance $x-y+z=3$, $x+y+z=4$ is

(1) $\cos^{-1}\frac{1}{3}$ (2) $\cos^{-1}\frac{4}{41}$

(3) $\sin^{-1}\frac{4}{27}$ (4) $\sin^{-1}\frac{4}{41}$

47. If the vectors $\bar{i}-2x\bar{j}+3y\bar{k}$ and $\bar{i}-2x\bar{j}-3y\bar{k}$ are perpendicular then the locus of (x,y) is :

(1) A circle (2) An ellipes

(3) A parabola (4) A hyperbola

Applications of Scalar Product in Mechanics to find the work done:

48. The force $\bar{F}=3\bar{i}+\bar{j}-\bar{k}$ acts on a particle and it moves from the point $A(2\bar{i}-\bar{j})$ to $B(2\bar{i}+\bar{j})$. The work done by the force $\bar{F}=$

1) 1 2) 2 3) 3 4) 4

49. The work done by force $\bar{F}=a\bar{i}+\bar{j}+\bar{k}$ in moving a particle from $(1,1,1)$ to $(2,2,2)$ along a stright line is 5 units. Then $a=$

(1) 1 (2) 2 (3) 3 (4) 4

LEVEL - I (H.W)-KEY

01) 3	02) 2	03) 1	04) 4	05) 2	06) 1
07) 3	08) 4	09) 1	10) 3	11) 2	12) 1
13) 3	14) 3	15) 2	16) 2	17) 4	18) 4
19) 2	20) 3	21) 1	22) 1	23) 1	24) 4
25) 3	26) 3	27) 3	28) 1	29) 3	30) 1
31) 1	32) 1	33) 3	34) 3	35) 2	36) 3
37) 2	38) 3	39) 3	40) 3	41) 2	42) 2
43) 4	44) 4	45) 1	46) 1	47) 4	48) 2
49) 3					

LEVEL - I (H.W)-HINTS

1. $\frac{\bar{b}.\bar{a}}{|\bar{a}|}$

2. $\frac{(\bar{a}.\bar{b})\bar{b}}{|\bar{b}|^2}$.

3. $\frac{|(3\bar{a}-2\bar{b}).\bar{c}|}{|\bar{c}|}=3$.

4. $\frac{\bar{a}.\bar{b}}{|\bar{b}|}=\frac{-6}{\sqrt{3}}\Rightarrow\frac{|\bar{a}||\bar{b}|\left(-\frac{\sqrt{3}}{2}\right)}{|\bar{b}|}=\frac{-6}{\sqrt{3}}\Rightarrow|\bar{a}|=4$.

5. $\frac{(a\bar{i}+b\bar{j}+c\bar{k}).\bar{j}}{|\bar{j}|}=b$.

6. $\sqrt{a^2+b^2+c^2}=\sqrt{4+1+4}=3$

7. $\dfrac{3\left(\bar{i}+\bar{j}+\bar{k}\right)}{\sqrt{3}}$

8. $\dfrac{\left|\bar{a}.\bar{b}\right|}{\left|\bar{b}\right|}$.

9. $\bar{a}+\bar{b}=4\bar{i}-6\bar{j}+9\bar{k}, 2\bar{a}-\bar{b}=\bar{i}+3\bar{j}$

$\left(2\bar{a}-\bar{b}\right).\left(\bar{a}+\bar{b}\right)=-4-18=-22$

$=\dfrac{\left|\left(2\bar{a}-\bar{b}\right).\left(\bar{a}+\bar{b}\right)\right|}{\left|\bar{a}+\bar{b}\right|}=\dfrac{22}{\sqrt{133}}$

10. The projection of $\bar{b}$ along $\bar{a}$ $=\left(\dfrac{\left(\bar{a}.\bar{b}\right)}{\left|\bar{a}\right|^2}\right)\bar{a}$

11. Component of $\bar{a}$ perpendicular to the direction of

$\bar{a}=\bar{a}-\dfrac{\left(\bar{a}.\bar{b}\right)}{\left|\bar{b}\right|^2}\bar{b}$

12. $\dfrac{\left|\left(3\bar{a}-2\bar{b}\right).\bar{c}\right|}{\left|\bar{c}\right|}=3$

13. Here $\bar{r}.\left(\bar{a}-\bar{b}\right)=\bar{c}.\left(\bar{a}-\bar{b}\right)\Rightarrow\bar{r}-\bar{c}$ perpendicular to $\bar{a}-\bar{b}\Rightarrow\overline{CP}$ Perpendicular to $\overline{AB}$. Similarly $\overline{AP}$ Perpendicular to $\overline{AB}\Rightarrow\overline{CP}$ *and* $\overline{AP}$ are altitudes $\therefore 'P'=$ Ortho Center of ΔABC.

14. $\left|\bar{a}+\bar{b}+\bar{c}\right|^2=0\Rightarrow 1+1+\left|\bar{c}\right|^2-3=0 \Rightarrow\left|\bar{c}\right|=1$

15. $\left|\bar{a}+\bar{b}+\bar{c}\right|^2=0 \Rightarrow\left|\bar{a}\right|^2+\left|\bar{b}\right|^2+\left|\bar{c}\right|^2+$

$2\left(\bar{a}.\bar{b}+\bar{b}.\bar{c}+\bar{c}.\bar{a}\right)=0$

16. $\left(\bar{i}+\bar{j}+\bar{k}\right).\dfrac{\left((2+x)\bar{i}+6\bar{j}-2\bar{k}\right)}{\left|(2+x)\bar{i}+6\bar{j}-2\bar{k}\right|}=1$

17. $\left.\begin{array}{l}\bar{r}.\bar{i}=3....(1)\\ \bar{r}.\bar{j}=5....(2)\end{array}\right\}$ Solving (1) & (2).

18. Use orthogonal projection formula.

19. Dot product $=0$.

20. $\left|\bar{a}+\bar{b}\right|^2+\left|\bar{a}-\bar{b}\right|^2=2\left(\left|\bar{a}\right|^2+\left|\bar{b}\right|^2\right)$

$\Rightarrow 1+49=2\left(16+\left|\bar{b}\right|^2\right)\Rightarrow 2\left|\bar{b}\right|^2=18 \Rightarrow\left|\bar{b}\right|=3$.

21. Let $\bar{a}=x\bar{i}+y\bar{j}+z\bar{k}$ then $x=1, y=0, z=0$ so $\bar{a}=\bar{i}$

22. Check from options that $\bar{b}=\lambda\bar{a}$ and $\bar{a}.\bar{b}=3$.

23. Use $\bar{r}=\left(\bar{r}.\bar{a}\right)\bar{a}+\left(\bar{r}.\bar{b}\right)\bar{b}+\left(\bar{r}.\bar{c}\right)\bar{c}$.

24. $|\bar{a}+\bar{b}|<1$ squaring on both sides.

25. Conceptual

26. Expand.

27. $\cos\theta=\dfrac{\bar{a}.\bar{b}}{\left|\bar{a}\right|\left|\bar{b}\right|}$

28. $\cos\theta=\dfrac{\bar{a}.\bar{b}}{|\bar{a}||\bar{b}|}$

29. Here $\bar{\alpha}$ perpendicular $\bar{\beta}\Rightarrow\left(\bar{\alpha},\bar{\beta}\right)=\dfrac{\pi}{2}$

$\Rightarrow\left(-\bar{\alpha},\bar{\beta}\right)=\pi-\dfrac{\pi}{2}=\dfrac{\pi}{2}$

30. Find $\cos\left(\overline{AB},\bar{i}\right)$, $\cos\left(\overline{AB},\bar{j}\right)$, $\cos\left(\overline{AB},\bar{k}\right)$.

31. $(x,3,-7).(x,-x,4)<0$.

32. $\left|\bar{a}\right|^2=\left|\bar{b}+\bar{c}\right|^2 \Rightarrow a^2=b^2+c^2+2\bar{b}.\bar{c}$

$a^2=b^2+c^2\left[\because\bar{b}.\bar{c}=0\right]$.

33. $\cos A=\dfrac{\overline{AB}.\overline{AC}}{\left|\overline{AB}\right|\left|\overline{AC}\right|}$.

34. Check $\left(\bar{a}+\bar{b}\right).\left(\bar{a}-\bar{b}\right)=0$ (or) check $\left|\bar{a}\right|=\left|\bar{b}\right|$.

36. $\left|\bar{a}-\bar{b}\right|^2=\left|\bar{a}\right|^2+\left|b\right|^2-2\left(\bar{a}.\bar{b}\right)=34+15=49$

Therefore, $\left|\bar{a}-\bar{b}\right|=7$

36. $\bar{a}.\bar{b}=12-6-2=4$ $|\bar{a}|=3,|\bar{b}|=7$

$\therefore \cos\theta=\frac{\bar{a}.\bar{b}}{|\bar{a}||\bar{b}|}=\frac{4}{21}$

37. $(e_1+2e_2).(5e_1-4e_2)=0 \Rightarrow 5-8+6\cos\theta$

where, $\theta=(e_1,e_2)\Rightarrow\cos\theta=\frac{1}{2}\Rightarrow\theta=60^0$

38. $\bar{a}+\bar{b}=-\sqrt{3}\bar{c}\Rightarrow(\bar{a}+\bar{b})^2=3c^2$

$\Rightarrow 1+1+2(\bar{a}.\bar{b})=3\Rightarrow\cos\theta=\frac{1}{2}\quad\Rightarrow\theta=60^0$

39. $\cos\theta=\frac{a.b}{|a||b|}$

40. $a.b=2|a|^2\cos120^0=-|a|^2$

$(a-xb).(a-b)=0\quad\Rightarrow x=\frac{2}{5}$

41. $\theta=\cos^{-1}\left(\frac{\overline{BD}.\overline{AC}}{|\overline{BD}||\overline{AC}|}\right)$

42. $(\bar{r}-\bar{a}).\bar{b}=0$

$\Rightarrow\left[(x-3)\bar{i}+(y+2)\bar{j}+(z-1)\bar{k}\right].$

$(4\bar{i}+7\bar{j}-4\bar{k})=0.$

43. Distance $=\frac{|d|}{\sqrt{a^2+b^2+c^2}}.$

44. Distance $=\frac{|d|}{\sqrt{a^2+b^2+c^2}}.$

45. $\cos\theta=\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}.$

46. $\cos\theta=\frac{a_1a_2+b_1b_2+c_1c_2}{\sqrt{a_1^2+b_1^2+c_1^2}\sqrt{a_2^2+b_2^2+c_2^2}}.$

47. Take $\bar{a}.\bar{b}=0$

48. $W=\bar{F}.\overline{AB}$

49. Work done = 5 $\Rightarrow\bar{F}.\bar{S}=5\Rightarrow a=3$

LEVEL - II (C.W)

Vectors Geometrical Interpretation Orthogonal Projection:

1. If $\bar{a}=2\bar{i}+\bar{j}+2\bar{k}$,$\bar{b}=5\bar{i}-3\bar{j}+\bar{k}$, then the length of the component vector of b perpendicular to a is

1) $\sqrt{13}$ 2) $\sqrt{26}$ 3) $\sqrt{18}$ 4) $\sqrt{20}$

2. A Parallelogram is constructed with $\bar{a}$ and $\bar{b}$ as adjacent sides such that $|\bar{a}|=a$ and $|\bar{b}|=b$. The vector which coincides with the altitude of the parallelogram and is perpendicular to the vector $\bar{a}$ is.

1) $\bar{b}-\frac{(\bar{a}.\bar{b})\bar{a}}{a^2}$ 2) $\bar{a}-\frac{(\bar{a}.\bar{b})\bar{b}}{b^2}$

3) $\bar{a}-\frac{(\bar{a}.\bar{b})\bar{b}}{a^2}$ 4) $\bar{b}-\frac{(\bar{a}.\bar{b})\bar{a}}{b^2}$

3. If $\bar{a},\bar{b},\bar{c}$ are position vectors of the non-collinear points A, B, C respectively, the shortest distance of A from BC is

1) $\bar{a}.(\bar{b}-\bar{c})$ 2) $\bar{b}.(\bar{c}-\bar{a})$

3) $|\bar{b}-\bar{a}|$ 4) $\sqrt{|\bar{b}-\bar{a}|^2-\left(\frac{(\bar{a}-\bar{b}).(\bar{c}-\bar{b})}{|\bar{c}-\bar{b}|}\right)^2}$

Properties of DOT Product:

4. If $\bar{a}$ and $\bar{b}$ are non-collinear unit vectors and $|\bar{a}+\bar{b}|=\sqrt{3}$ then $(2\bar{a}+5\bar{b}).(3\bar{a}-\bar{b})=$

1) $\frac{15}{4}$ 2) $\frac{15}{2}$ 3) 15 4) 16

5. If $\bar{a},\bar{b}$ and $\bar{c}$ are perpendicular to $\bar{b}+\bar{c},\bar{c}+\bar{a}$ and $\bar{a}+\bar{b}$ respectively and if $|\bar{a}+\bar{b}|=6$, $|\bar{b}+\bar{c}|=8$ and $|\bar{c}+\bar{a}|=10$, then $|\bar{a}+\bar{b}+\bar{c}|$ is equal to

1) $5\sqrt{2}$ 2) 50 3) $10\sqrt{2}$ 4) 10

6. If $\bar{b}=4\bar{i}+3\bar{j}$ and c are two vectors perpendicular

to each other in the XY plane ,the vector in the same plane having components 1, 2 along b and c respectively is

1) $(-2\bar{i}+11\bar{j})/5$ 2) $(2\bar{i}+11\bar{j})/5$

3) $(-2\bar{i}-11\bar{j})/5$ 4) $(2\bar{i}-11\bar{j})/5$

7. **The perpendicular distance of a corner of unit cube from a diagonal not passing through it is**

1) $\sqrt{2/3}$ 2) 2/3 3) 1/3 4) 1

8. **If A, B are two points on the curve $y = x^2$ in the xoy plane satisfying $\overline{OA}.\bar{i} = 1$ and $\overline{OB}.\bar{i} = -2$ then the length of the vector $2\overline{OA} - 3\overline{OB}$ is**

1) $\sqrt{14}$ 2) $2\sqrt{51}$ 3) $3\sqrt{41}$ 4) $2\sqrt{41}$

9. $\bar{a}$ **&** $\bar{b}$ **are two non-collinear vectors then the value of** $\left\{\frac{\bar{a}}{|\bar{a}|^2} - \frac{\bar{b}}{|\bar{b}|^2}\right\}^2$ **is equal to**

1) $\frac{\bar{a}-\bar{b}}{ab}$ 2) $\left(\frac{\bar{a}-\bar{b}}{ab}\right)^2$

3) $\left(\frac{\bar{a}-\bar{b}}{ab}\right)^3$ 4) $\left(\bar{a}-\bar{b}\right)^2$

10. **Let $\bar{u} = \bar{i}+\bar{j}, \bar{v} = \bar{i}-\bar{j}$ and $\bar{w} = \bar{i}+2\bar{j}+3\bar{k}$. If $\bar{n}$ is a unit vector such that $\bar{u}.\bar{n} = 0$ and $\bar{v}.\bar{n} = 0$, then $|\bar{w}.\bar{n}| =$** **(AIE-2003)**

1) 1 2) 2 3) 3 4) 0

11. **If a, b, c are the p^{th}, q^{th}, r^{th} terms of an HP and**

$\bar{u} = (q-r)\bar{i} + (r-p)\bar{j} + (p-q)\bar{k}$

$\bar{v} = \frac{\bar{i}}{a} + \frac{\bar{j}}{b} + \frac{\bar{k}}{c}$ **then**

1) $\bar{u}, \bar{v}$ are parallel vectors

2) $\bar{u}, \bar{v}$ are orthogonal vectors

3) $\bar{u}.\bar{v} = 1$ 4) $\bar{u} \times \bar{v} = \bar{i} + \bar{j} + \bar{k}$

12. **If a parallelogram is constructed on the vectors $\bar{a} = 3\bar{p} - \bar{q}, \bar{b} = \bar{p} + 3\bar{q}$ and $|\bar{p}| = |\bar{q}| = 2$ and angle between $\bar{p}$ and $\bar{q}$ is $\frac{\pi}{3}$, then the ratio of the lengths of the sides is**

1) $\sqrt{7} : \sqrt{13}$ 2) $\sqrt{6} : \sqrt{2}$

3) $\sqrt{3} : \sqrt{5}$ 4) $1 : 2$

Angle between two Vectors:

13. **If $\bar{a} = 2\bar{m} + \bar{n}, \bar{b} = \bar{m} - 2\bar{n}$, Angle between the unit vectors m and n is 60^0 . a,b are the sides of a parallelogram ,then the lengths of the diagonals are**

1) $\sqrt{7}, \sqrt{5}$ 2) $\sqrt{13}, \sqrt{5}$

3) $\sqrt{7}, \sqrt{13}$ 4) $\sqrt{11}, \sqrt{13}$

14. **In a right angled ΔABC. the hypotenuse AB = p then AB.AC+BC.BA+CA.CB is**

1) $2p^2$ 2) $p^2/2$ 3) p^2 4) 0

15. **The triangle ABC is defined by the vertices $A = (0,7,10), B = (-1,6,6)$ and $C = (-4,9,6)$. Let D be the foot of the altitude from B to the side AC Then $\overline{DB} =$**

1) $\bar{i} + 2\bar{j} + 2\bar{k}$ 2) $\bar{i} - 2\bar{j} - 2\bar{k}$

3) $\bar{i} + 2\bar{j} - 2\bar{k}$ 4) $\bar{i} - 2\bar{j} + 2\bar{k}$

Vector Eqution of a Plane-Normal Form:

16. **A plane is at a distance of 8 units from the origin and is perpendicular to the vector $2\bar{i} + \bar{j} + 2\bar{k}$ then the equation of the plane is**

1) $\bar{r}.(2\bar{i} + \bar{j} + 2\bar{k}) = 8$ 2) $\bar{r}.(\bar{i} + 2\bar{j} + 2\bar{k}) = 24$

3) $\bar{r}.(2\bar{i} + 2\bar{j} + \bar{k}) = 24$ 4) $\bar{r}.(2\bar{i} + \bar{j} + 2\bar{k}) = 24$

17. **The angle between the planes passing through the points A(0, 0, 0), B(1, 1, 1) , C(3, 2, 1) & the planes passing through A(0, 0, 0),B(1, 1, 1), D(3, 1, 2) is**

1) 90^0 2) 45^0 3) 120^0 4) 30^0

Application of Scalar Products in Mechanics to find the work done:

18. **The point of application of the force (-2,4,7) is displaced from the point (3,-5,1) to the point (5,9,7) .But the force is suddenly halved when**

the point of application moves half the distance. The work done by the force is
1) 70 2) 70.5 3) 75 4) 75.5

19. **If forces of magnitudes 6 and 7 units acting in the directions $\bar{i}-2\bar{j}+2\bar{k}$ and $2\bar{i}-3\bar{j}-6\bar{k}$ respectively act on a particle which is displaced from the point $P(2,-1,-3)\ to\ Q(5,-1,1)$, then the work done by the forces is**
1) 4 units 2) -4 units 3) 7 units 4) -7 units

20. **A constant force $3\bar{i}+4\bar{j}-5\bar{k}$ acts on a particle at $\bar{i}+2\bar{j}+2\bar{k}$ and moves it to a point on the z-axis which is 3 units from origin, the work done is**
1) 16 2) -16 3) 14 4) -14

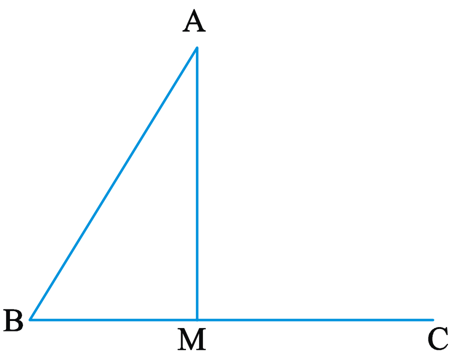
LEVEL-II (C.W)-KEY

01) 2	02) 1	03) 4	04) 2	05) 4	06) 1
07) 1	08) 4	09) 2	10) 3	11) 2	12) 1
13) 3	14) 3	15) 2	16) 4	17) 3	18) 2
19) 1	20) 2				

LEVEL-II (C.W)-HINTS

1. $\bar{b}-\dfrac{(\bar{a}.\bar{b})\bar{a}}{|\bar{a}|^2}$

2. Required vector = Component vector of $\bar{b}$ perpendicular to $\bar{a}$ = $\bar{b}-\dfrac{(\bar{b}.\bar{a})\bar{a}}{|\bar{a}|^2}$

3. 

$BM=\dfrac{\overline{BA}.\overline{BC}}{|\overline{BC}|}$, $AM=\sqrt{AB^2-BM^2}$

4. $|\bar{a}+\bar{b}|=\sqrt{3}\Rightarrow \bar{a}.\bar{b}=\dfrac{1}{2}$

Find $(2\bar{a}+5\bar{b}).(3\bar{a}-\bar{b})=\dfrac{15}{2}$

5. $|\bar{a}+\bar{b}|^2+|\bar{b}+\bar{c}|^2+|\bar{c}+\bar{a}|^2=200$

$\Rightarrow 2\left(|\bar{a}|^2+|\bar{b}|^2+|\bar{c}|^2\right)=200\left(\because \bar{a}.\bar{b}+\bar{b}.\bar{c}+\bar{c}.\bar{a}=0\right)$

$|\bar{a}+\bar{b}+\bar{c}|=\sqrt{100}=10$

6. Let $\bar{d}=x\bar{i}+y\bar{j}$, and $\bar{c}=-3\bar{i}+4\bar{j}$

$\dfrac{\bar{d}.\bar{b}}{|\bar{b}|}=1\Rightarrow 4x+3y=5,\ \dfrac{\bar{d}.\bar{c}}{|\bar{c}|}=2$

$\Rightarrow -3x+4y=10$

Solve these two equations.

7. The perpendicular distance of a corner of unit cube from a diagonal not passing through it is $\sqrt{\dfrac{2}{3}}$.

8. Let $A(x_1,y_1)\ \&\ B(x_2,y_2)$ lie on the parabola so that $\overline{OA}=\bar{i}+\bar{j}$, $\overline{OB}=-2\bar{i}+4\bar{j}$

9. Conceptual

10. Let $\bar{n}=a_1\bar{i}+a_2\bar{j}+a_3\bar{k}$

$\therefore\ \bar{u}.\bar{n}=0\ \Rightarrow a_1+a_2=0$

$\bar{v}.\bar{n}=0\ \Rightarrow a_1-a_2=0$

$\Rightarrow a_1=0,\ a_2=0$

Also, $a_3=1$ ($\because \bar{n}$ is a unit vector)

$\therefore |\bar{w}.\bar{n}|=|\bar{w}||\bar{n}|=3$.

11. Use $\dfrac{1}{a}=A+(P-1)D\ \&\ q-r=\dfrac{1}{b}-\dfrac{1}{c}=\dfrac{c-b}{bc}$ etc

12. Find $|\bar{a}|=\sqrt{9p^2+q^2-6pq\cos 60^o}=2\sqrt{7}$;

$|\bar{b}|=\sqrt{p^2+9q^2+6pq\cos 60^o}=2\sqrt{13}$

13. Length of the diagonals are $|\bar{a}+\bar{b}|,|\bar{a}-\bar{b}|$

14. $\overline{AB}.\overline{AC}+\overline{BC}.\overline{BA}+\overline{CA}.\overline{CB}$
expand and simplify.

15. Check from options that $\overline{AC}.\overline{BD}=0$

16. $\bar{r}.\hat{n} = p \Rightarrow \bar{r}.\left(\frac{2\bar{i}+\bar{j}+2\bar{k}}{3}\right) = 8$

$\Rightarrow \bar{r}.\left(2\bar{i}+\bar{j}+2\bar{k}\right) = 24$

17. Plane through ABC is $-x+2y-z=0$

plnae through ABD is $x+y+-2z=0$

$\overline{n_1} = -\bar{i}+2\bar{j}-\bar{k}$, $\overline{n_2} = \bar{i}+\bar{j}-2\bar{k}$

$\cos\theta = \frac{\overline{n_1}.\overline{n_2}}{|\overline{n_1}||\overline{n_2}|} = \frac{1}{2} \Rightarrow \theta = 60^0\ (or)\ 120^0$

18. $\bar{F}.\frac{\bar{d}}{2} + \frac{\bar{F}}{2}.\frac{\bar{d}}{2}$

19. $\bar{F} = \bar{F_1}+\bar{F_2} = (4,-7,-2), \bar{d} = \overline{AB} = (3,0,4)$

$W.D. = \bar{F}.\bar{d} = 4$

20. $W = \bar{F}.\overline{AB}$

$=\left(3\bar{i}+4\bar{j}-5\bar{k}\right)\left(3\bar{k}-(\bar{i}+2\bar{j}+2\bar{k})\right)$

$=\left(3\bar{i}+4\bar{j}-5\bar{k}\right).\left(-\bar{i}-2\bar{j}+\bar{k}\right)$ = -16.

LEVEL - II (H.W)

Vectors Geometrical Interpretation Orthogonal Projection:

1. The vector component of $\bar{i}+\bar{j}+\bar{k}$ perpendicular to the vector $2\bar{i}-\bar{j}+2\bar{k}$ is

1) $\frac{\bar{i}+4\bar{j}+\bar{k}}{3}$ 2) $\frac{\bar{i}-4\bar{j}-3\bar{k}}{4}$

3) $\bar{i}+\bar{j}-\frac{1}{2}\bar{k}$ 4) $\bar{0}$

2. If $\bar{a}=3\bar{i}+\bar{j}+2\bar{k}$ and $\bar{b}=\bar{i}+2\bar{j}+3\bar{k}$ then a unit vector in the direction of the resultant of orthogonal projection of $\bar{b}$ on $\bar{a}$ and the projection of $\bar{b}$ on a line perpendicular to $\bar{a}$ is

1) $\frac{\bar{i}+2\bar{j}+3\bar{k}}{\sqrt{14}}$ 2) $\frac{2\bar{i}+\bar{j}+3\bar{k}}{\sqrt{14}}$

3) $\frac{3\bar{i}+\bar{j}+2\bar{k}}{\sqrt{14}}$ 4) $\frac{\bar{i}+3\bar{j}+2\bar{k}}{\sqrt{14}}$

3. The projection of the vector $\bar{i}+\bar{j}+\bar{k}$ on the line whose vectro equation is $\bar{r}=(3+t)\bar{i}+(2t-1)\bar{j}+3t\bar{k}$, t being the scalar parameter, is

(1) $\frac{1}{\sqrt{14}}$ (2) 6 (3) $\frac{6}{\sqrt{14}}$ (4) 10

Properties of DOT Product:

4. Let $\bar{u},\bar{v},\bar{w}$ be such that $|\bar{u}|=1, |\bar{v}|=2, |\bar{w}|=3$. If the projection of $\bar{v}$ along $\bar{u}$ is equal to that of $\bar{w}$ along $\bar{u}$ and $\bar{v},\bar{w}$ are perpendicular to each other, then $|\bar{u}-\bar{v}+\bar{w}|$ equals

1) 2 2) $\sqrt{7}$ 3) $\sqrt{14}$ 4) 14

5. If $\bar{a}+\bar{b}+\bar{c}=0$, then $\bar{a}.\bar{a}+\bar{b}.\bar{b}+2\bar{a}.\bar{b}-\bar{c}.\bar{c}=$

1) $2\left(\bar{b}.\bar{c}+\bar{c}.\bar{a}\right)$ 2) 0

3) Null vector 4) $-2\left(\bar{b}.\bar{c}+\bar{c}.\bar{a}\right)$

6. The length of longer diagonal of the parallelogram constructed on $5\bar{a}+2\bar{b}$ and $\bar{a}-3\bar{b}$ if it is given $|\bar{a}|=2\sqrt{2}; |\bar{b}|=3$, and $\left(\bar{a},\bar{b}\right)=\pi/4$ is

1) 15 2) $\sqrt{113}$ 3) $\sqrt{593}$ 4) $\sqrt{395}$

7. If $\bar{a},\bar{b},\bar{c}$ are three vectors such that each is inclined at an angle $\frac{\pi}{3}$ with the other two and $|\bar{a}|=1, |\bar{b}|=2, |\bar{c}|=3$ then the scalar product of the vectors $2\bar{a}+3\bar{b}-5\bar{c}$ and $4\bar{a}-6\bar{b}+10\bar{c}$ is

1) 188 2) -334 3) -522 4) -514

Angle between two Vectors:

8. If $\bar{a},\bar{b},\bar{c}$ are three mutually perpendicular vectors of equal magnitudes then the vector

equally inclined to $\bar{a},\bar{b},\bar{c}$

1) $\frac{\bar{i}+\bar{j}+\bar{k}}{\sqrt{3}}$ 2) $\bar{a}-\bar{b}+\bar{c}$

3) $\frac{\bar{a}-\bar{b}-\bar{c}}{\sqrt{3}}$ 4) $\frac{\bar{a}+\bar{b}+\bar{c}}{\sqrt{3}}$

9. If the position vectors of A,B,C,D are respectively $(-6,1,6)(6,-2,3),(-2,-3,-1)$ *and* $(-5,-9,-7)$ then

1) $\lfloor BCA$ is a right angle 2) $\lfloor CDA$ is a right angle

3) $\lfloor ABD$ is a right angle 4) $\lfloor ACD$ is a right angle

10. Let $A(2,-3,0)B(2,-1,1)C(0,1,4)$ be the vertices of ΔABC. Then angle between the median $\overline{BD}$ and the base $\overline{AC}$ is

1) 30^0 2) 45^0 3) 60^0 4) 90^0

11. Consider points A, B, C, D with position vectors $7\bar{i}-4\bar{j}+7\bar{k},\bar{i}-6\bar{j}+10\bar{k},-\bar{i}-3\bar{j}+4\bar{k}$ and $5\bar{i}-\bar{j}+5\bar{k}$ respectively. Then, ABCD is a (AIE-2003)

1) rhombus 2) rectangle
3) parallelogram but not a rhombus
4) None of these

12. If ABCDEF is a regular hexagon, then $\overline{AB}.\overline{AF}=...$

1) $\frac{1}{2}BC^2$ 2) $-\frac{1}{2}BC^2$ 3) $\frac{1}{2}AC^2$ 4) $-\frac{1}{2}AC^2$

13. $|\bar{b}|=6$, then $\bar{b}-3\bar{c}=\lambda\bar{a}$ if $\lambda=$

1) $-9,3$ 2) $-3,6$ 3) 6, 3 4) -3, 4

14. The vectors a,b,c are of same length and taken pairwise they form equal angles, If $\bar{a}=\bar{i}+\bar{j},\bar{b}=\bar{j}+\bar{k}$ then $\bar{c}$ =

1) $\bar{i}+\bar{k}$ 2) $\bar{i}+\bar{j}+\bar{k}$ 3) $\bar{i}-\bar{j}$ 4) $-\bar{i}+\bar{j}$

15. If $\bar{a}=-\bar{i}+\bar{j}+\bar{k}$ and $\bar{b}=2\bar{i}+\bar{k}$, then the vector c satisfying the conditions that (i) it is coplanar with a and b (ii) it is perpendicular to b (iii) a.c =7 is

1) $-3/2\bar{i}+5/2\bar{j}+3\bar{k}$ 2) $-3\bar{i}+5\bar{j}+6\bar{k}$

3) $-6\bar{i}+\bar{k}$ 4) $-\bar{i}+2\bar{j}+2\bar{k}$

16. If $a+2b+3c=4$ then the least value of $a^2+b^2+c^2$

1) $\frac{2}{7}$ 2) $\frac{3}{7}$ 3) $\frac{5}{7}$ 4) $\frac{8}{7}$

17. Angle between $\bar{a}$ & $\bar{b}$ is 120^0. If $|\bar{b}|=2|\bar{a}|$ and the vectors $\bar{a}+x\bar{b},\bar{a}-\bar{b}$ are right angles. Then x is equal to

1) $\frac{1}{3}$ 2) $\frac{1}{5}$ 3) $\frac{2}{5}$ 4) $\frac{2}{3}$

18. If $\bar{a}=\bar{i}+\bar{j}+\bar{k}$ *and* $\bar{b}=\bar{i}-\bar{j}$ then the vectors $(\bar{a}.\bar{i})\bar{i}+(\bar{a}.\bar{j})\bar{j}+(\bar{a}.\bar{k})\bar{k},(\bar{b}.\bar{i})\bar{i}+(\bar{b}.\bar{j})\bar{j}+(\bar{b}.\bar{k})\bar{k}$ and $\bar{i}+\bar{j}-2\bar{k}$

1) Are mutually Perpendicular
2) Coplanar
3) Forms a Parallelopiped of volume 2 units
4) Forms a Parallelopiped of volume 3 units

19. $\bar{a},\bar{b},\bar{c}$ are the position vectors of the vertices of a triangle ABC ,If $(\bar{a}-\bar{b}).(\bar{c}-\bar{b})=0$,then the position vector of circumcentre is

1) $\frac{(\bar{b}+\bar{c})}{2}$ 2) $\frac{(\bar{a}+\bar{b})}{2}$

3) $\frac{(\bar{a}+\bar{c})}{2}$ 4) $\frac{\bar{a}+\bar{b}+\bar{c}}{2}$

Applications of scalar Product in Mechanics to find the work done:

20. Constant forces $\bar{P}=2\bar{i}-5\bar{j}+6\bar{k}$ and $\bar{Q}=-\bar{i}+2\bar{j}-\bar{k}$ act on a particle, determine the work done when the particle is displaced from the point A with position vector $4\bar{i}-3\bar{j}-2\bar{k}$ to a point B with position vector $6\bar{i}+\bar{j}-3\bar{k}$

1) 10 2) -15 3) -10 4) 15

LEVEL - II (H.W)-KEY

01) 1 02) 1 03) 3 04) 3 05) 2 06) 3

07) 2 08) 4 09) 1 10) 2 11) 4 12) 2
13) 1 14) 1 15) 1 16) 4 17) 3 18) 1
19) 3 20) 2

LEVEL - II (H.W)-HINTS

1. Use $\bar{b}-\frac{(\bar{a}.\bar{b})\bar{a}}{|\bar{a}|^2}$.

2. Required vector $=\frac{\bar{b}}{|\bar{b}|}$.

3. Since $\bar{r}=3\bar{i}-\bar{j}+t(\bar{i}+2\bar{j}+3\bar{k})$

So, a vector parallel to the lines is

$\bar{b}=\bar{i}+2\bar{j}+3\bar{k}$

Now, unit vector along the line is

$\frac{\bar{i}+2\bar{j}+3\bar{k}}{\sqrt{I^2+2^2+3^2}}$ i.e $=\frac{\bar{i}+2\bar{j}+3\bar{k}}{\sqrt{14}}$ the projection of $(\bar{i}+\bar{j}+\bar{k})$ on lines is

$$(\bar{i}+\bar{j}+\bar{k}),\frac{(\bar{i}+2\bar{j}+3\bar{k})}{\sqrt{14}}=\frac{1+2+3}{\sqrt{14}}=\frac{6}{\sqrt{14}}$$

4. $|\bar{u}-\bar{v}+\bar{w}|^2=|\bar{u}|^2+|\bar{v}|^2+|\bar{w}|^2+2|\bar{u}.\bar{w}-\bar{u}.\bar{v}-\bar{v}.\bar{w}|$

$=1+4+9+2|\bar{u}.\bar{w}-\bar{u}.\bar{v}|$

$(\because \bar{v}.\bar{w}=0)=14+2(|\bar{u}||\bar{w}|\cos\alpha-|\bar{u}||\bar{v}|\cos\beta)$

But, $|\bar{v}|\cos\beta=|\bar{w}|\cos\alpha$

$\therefore|\bar{u}-\bar{v}+\bar{w}|^2=14\Rightarrow|\bar{u}-\bar{v}+\bar{w}|=\sqrt{14}$.

5. $\bar{a}+\bar{b}+\bar{c}=\bar{0}\Rightarrow\bar{a}+\bar{b}=-\bar{c}$

Take dot product on both sides with $\bar{a}$ and $\bar{b}$ then add.

6. Longer diagonal $-4\bar{a}-5\bar{b}$

its magnitude $=|-4\bar{a}-5\bar{b}|=\sqrt{593}$.

7. Given $(\bar{a},\bar{b})=(\bar{b},\bar{c})=(\bar{c},\bar{a})=60^o$

Find $(2\bar{a}+3\bar{b}-5\bar{c}).(4\bar{a}-6\bar{b}+10\bar{c})=-334$

8. $|\bar{a}|=|\bar{b}|=|\bar{c}|$ and $\bar{a}.\bar{b}=\bar{b}.\bar{c}=\bar{c}.\bar{a}=0$.

9. $\therefore\underline{|BCA}=90^0$.

10. $\theta=\cos^{-1}\left(\frac{\overline{BD}.\overline{AC}}{|\overline{BD}||\overline{AC}|}\right)$.

11. $\overline{AB}$ and $\overline{DC}$ are not parallel.

12. Drawn the figure and sec the $\underline{|BAF}=120^0.\therefore\overline{AB}.\overline{AF}=|\overline{AB}||\overline{AF}|\cos120^0$.

Here $AB=AF=a=BC$

13. Write $\bar{b}=\lambda\bar{a}+3\bar{c}$ squaring. we get

$\lambda^2\pm6\lambda-27=0$

14. $\bar{c}=\bar{i}+\bar{k}$

15. Take C $=\left(-\frac{3}{2},\frac{5}{2},3\right)$ and verify

16. Consider the vectors $\bar{p}=a\bar{i}+b\bar{j}+c\bar{k}$ and

$\bar{q}=\bar{i}+2\bar{j}+3\bar{k}$, $\cos\theta=\frac{\bar{p}.\bar{q}}{|\bar{p}||\bar{q}|}\le1$

17. $(\bar{a}+x\bar{b}).(\bar{a}-\bar{b})=0$

$\Rightarrow|\bar{a}|^2-(x-1)|\bar{a}|^2-4x|\bar{a}|^2=0$

(with $|\bar{b}|=2|\bar{a}|$ and $(\bar{a}.\bar{b})=120^0$)

$\Rightarrow(2-5x)|\bar{a}|^2=0\Rightarrow x=\frac{2}{5}(|\bar{a}|\ne0)$

18. Verification

19. ΔABC Right angled triangle at B

20. $W=\bar{F}.\bar{s},\bar{F}=\bar{P}+\bar{Q},\bar{S}=\overline{AB}$.

LEVEL - III

1. ^{n} $\overline{a}_i| = 1$, for all i, then the value of $\sum\limits_{1 \le i} \sum\limits_{< j \le n} \overline{a}_i . \overline{a}_j$ is

1) n^2 2) $-n^2$ 3) n 4) $-\frac{n}{2}$

2. The value of a for which the angle between $\overline{a} = 2a^2\overline{i} + 4a\overline{j} + \overline{k}$ and $\overline{b} = 7\overline{i} - 2\overline{j} + a\overline{k}$ is obtuse and the angle between $\overline{b}$ and z-axis is acute and less than $\frac{\pi}{6}$ is

1) Does not exist 2) Lies in $\left(0, \frac{1}{2}\right)$

3) Lies in $(-1,1)$ 4) Lies in $(0,1)$

3. If the position vector of a point P is $\overline{r} = x\overline{i} + y\overline{j} + z\overline{k}$, where $x, y, z \in N$ and $\overline{\alpha}$ is a vector given by $\overline{\alpha} = \overline{i} + \overline{j} + \overline{k}$, then the total number of possible positions of point P for which $\overline{r}.\overline{\alpha} = 10$

1) 36 2) 72 3) 66 4) 100

4. Let $\overline{a} = 2\overline{i} + \overline{j} + \overline{k}, \overline{b} = \overline{i} + 2\overline{j} - \overline{k}$ and a unit vector $\overline{c}$ be coplanar. If $\overline{c}$ is perpendicular to $\overline{a}$, then $\overline{c}$ is

1) $\frac{1}{\sqrt{2}}\left(-\overline{j} + \overline{k}\right)$ 2) $\frac{1}{\sqrt{3}}\left(-\overline{i} - \overline{j} - \overline{k}\right)$

3) $\frac{1}{\sqrt{5}}\left(\overline{i} - 2\overline{j}\right)$ 4) $\frac{1}{\sqrt{5}}\left(\overline{i} - \overline{j} - \overline{k}\right)$

5. The position vector of the foot of the perpendicular from (1,-2,-3) to the line $\overline{r} = \overline{i} + \overline{j} + \lambda\left(2\overline{i} + \overline{j} + \overline{k}\right)$ is

1) $-2\overline{i} - \overline{j} + \overline{k}$ 2) $-\overline{i} - \overline{k}$

3) $\frac{\overline{j} + \overline{k}}{2}$ 4) $\overline{i} + \overline{j} + \overline{k}$

6. Let $\overline{a} = 2\overline{i} - \overline{j} + \overline{k}, \overline{b} = \overline{i} + 2\overline{j} - \overline{k}$ and $\overline{c} = \overline{i} + \overline{j} - 2\overline{k}$ be three vectors. A vector in the plane of $\overline{b}$ and $\overline{c}$ whose projection on $\overline{a}$ is of magnitude $\sqrt{\frac{2}{3}}$ is

1) $2\overline{i} - 3\overline{j} - 3\overline{k}$ 2) $2\overline{i} + 3\overline{j} + 3\overline{k}$

3) $-2\overline{i} - \overline{j} + 5\overline{k}$ 4) $2\overline{i} + \overline{j} + 5\overline{k}$

7. The ΔABC is defined by the vertices $A(1,-2,2)\, B(1,4,0)$ and $C(-4,1,1)$. Let M be the foot of the altitude drawn from the vertex B to side AC. Then $\overline{BM} =$

1) $\left(\frac{-20}{7}, \frac{-30}{7}, \frac{10}{7}\right)$ 2) $(-20,-30,10)$

3) $(2,3,-1)$ 4) $(1,2,3)$

8. If p^{th}, q^{th}, r^{th} terms of a G.P. are the positive numbers a, b, c then angle between the vectors $\log a^3\overline{i} + \log b^3\overline{j} + \log c^3\overline{k}$ and $(q-r)\overline{i} + (r-p)\overline{j} + (p-q)\overline{k}$ is

1) $\frac{\pi}{6}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ 4) $\sin^{-1}\left(\frac{1}{\sqrt{a^2 + b^2 + c^2}}\right)$

9. Let $\overline{a}, \overline{b}, \overline{c}$ be vectors of equal magnitude such that the angle between $\overline{a}$ and $\overline{b}$ is $\alpha, \overline{b}$ and $\overline{c}$ is β and $\overline{c}$ and $\overline{a}$ is γ. Then the minimum value of $\cos\alpha + \cos\beta + \cos\gamma$ is

1) $\frac{1}{2}$ 2) $-\frac{1}{2}$ 3) $\frac{3}{2}$ 4) $-\frac{3}{2}$

10. Let $\overline{\mathrm{p}}$ and $\overline{\mathrm{q}}$ be the position vectors of P and Q repectively with respect to 'O' and $|\mathbf{p}| = p, |\mathbf{q}| = q$. if R, S, divides PQ Internally and externally in the ratio 2:3 respectively. If OR and OS are perpendicular, then

1) $9p^2 = 4q^2$ 2) $4p^2 = 9q^2$

3) 9p=4q 4) 4p=9q

11. The vectors $\overline{X}$ and $\overline{Y}$ satisfy the equations $2\overline{X} + \overline{Y} = \overline{p}, \overline{X} + 2\overline{Y} = \overline{q}$ where $\overline{p} = \overline{i} + \overline{j}$ and $\overline{q} = \overline{i} - \overline{j}$. If θ is the angle

between $\overline{X}$ and $\overline{Y}$ then

1) $\cos\theta = \frac{4}{5}$ 2) $\sin\theta = \frac{1}{\sqrt{2}}$

3) $\cos\theta = -\frac{4}{5}$ 4) $\cos\theta = -\frac{3}{5}$

12. $A=(2,3,5), B=(-1,3,2)$ **and** $C=(\lambda,5,\mu)$ **are the vertices of a triangle. If the median AM is equally inclined to the coordinates axes, then**

1) $\lambda=10, \mu=7$ 2) $\lambda=-10, \mu=7$

3) $\lambda=7, \mu=10$ 4) $\lambda=-7, \mu=-10$

13. The vectors $3\overline{a}-5\overline{b}$ **and** $2\overline{a}+\overline{b}$ **are mutually perpendicular and the vectors** $\overline{a}+4\overline{b}$ **&** $-\overline{a}+\overline{b}$ **are also mutually perpendicular. Then the acute angle between** $\overline{a}$ **and** $\overline{b}$ **is**

1) $\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$ 2) $\pi-\cos^{-1}\left(\frac{19}{5\sqrt{43}}\right)$

3) $\cos^{-1}\left(\frac{9}{5\sqrt{43}}\right)$ 4) $\pi-\cos^{-1}\left(\frac{9}{5\sqrt{43}}\right)$

14. If $\overline{a}$ **is the position vector of A then the position vector of the foot of the perpendicular from A to the plane** $\overline{r}.\overline{b}=\overline{b}.\overline{c}$ **is.**

1) $\overline{b}+\frac{(\overline{c}-\overline{a})}{|\overline{b}|^2}\overline{b}$ 2) $\overline{a}+\frac{|(\overline{c}-\overline{a}).\overline{b}|\overline{b}}{|\overline{b}|^2}$

3) $\overline{b}+\frac{|(\overline{c}-\overline{b}).\overline{a}|}{|\overline{a}|^2}\overline{b}$ 4) $\overline{a}+\frac{|(\overline{c}-\overline{b}).\overline{a}|}{|\overline{a}|^2}\overline{b}$

15. Let $a=BC, b=CA, c=AB$ **be the sides of the triangle ABC. If G is the centroid of** ΔABC **such that** $\overline{GB}$ **and** $\overline{GC}$ **are inclined at an obtuse angle, then**

1) $5a^2>b^2+c^2$ 2) $5c^2>a^2+b^2$

3) $5b^2>a^2+c^2$ 4) None of these

16. The position vector of A is $p\overline{i}+\overline{j}$ **.If A is rotated about O through an angle** $\frac{\pi}{6}$ **in anti clock wise direction. It coincides with B whose position vector** $\overline{i}+q\overline{j}$ **.The value of p, q are**

1) $\sqrt{3},3$ 2) $\sqrt{3},\frac{1}{3}$

3) $\sqrt{3}, \sqrt{3}$ or $\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}$ 4) $\frac{1}{3},\frac{1}{3}$

17. In $\Delta ABC, |\overline{CB}|=a, |\overline{CA}|=b, |\overline{AB}|=c.$ **CD is median through the vertex C. Then** $\overline{CA}.\overline{CD}$ **equals**

1) $\frac{1}{4}(3a^2+b^2-c^2)$ 2) $\frac{1}{4}(a^2+3b^2-c^2)$

3) 4)

18. In triangle ABC if $\overline{AB}=\frac{\overline{u}}{|\overline{u}|}-\frac{\overline{v}}{|\overline{v}|}$ **and** $\overline{AC}=\frac{2\overline{u}}{|\overline{u}|}$ **where** $|\overline{u}|\neq|\overline{v}|$ **then**

1) 2)

3)

4)

19. In a parallelogram ABCD,

and =c , then has the value

1) 2)

3) 4)

20. If A is where on the curve . the tangent at A cuts the x-axis at B. The value of is

1) 2)

3) 140 4) none of these

21. Let ABCD be a parallelogram such that and be an acute angle. If is the vector that coincides with

the altitude directed from the vertex B to the side AD, then is given by

1.
2.
3.
4.

LEVEL-III-KEY

01) 4	02) 1	03) 1	04) 1
05) 2	06) 3	07) 1	08) 2
09) 4	10) 1	11) 3	12) 3
13) 1	14) 2	15) 1	16) 3
17) 2	18) 1	19) 1	20) 1
21) 2			

LEVEL-III-HINTS

1.

2.

,

There is no value satisfying

3.

No. of solutions =

4. Let and . But

5.

Let, ,

D.R'S of D.R'S of line

Pro of D.R'S =0

6. A vector in the plane of and is or

Its projection on

is

,

7. Here Component vector of

Perpendicular and

hence find

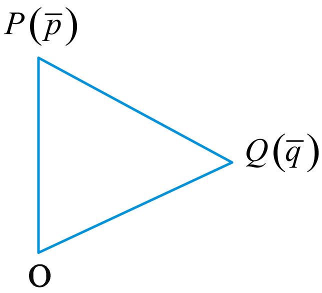
8. Use

etc.

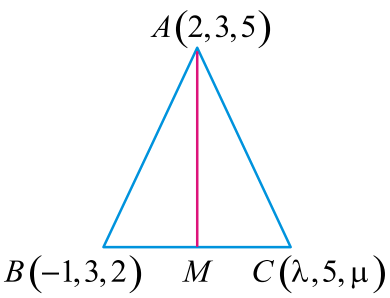
9. Let

we have,

Now,

10. 

11.

12. 

,

13.

from (1) & (2)

also

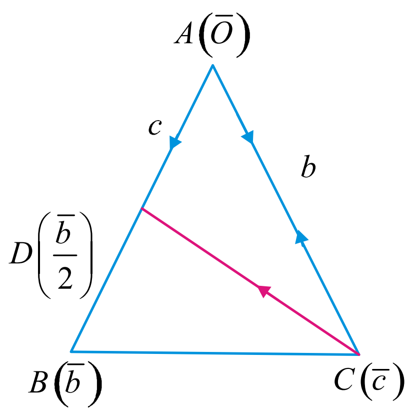
14. If then Component Vector of

along

15. Let then

Use

16. ,

17. 

Let A be origin & then

&

18.

Hence

19. or

in parallelogram

20. Putting in and

at A is 3. Tangent at A is .

it meets x-axis at B .

21. vector compenent of on

LEVEL - IV

1. **Let $\bar{a},\bar{b},\bar{c}$ be mutually perpendicular unit**

I :

II : $\bar{a}+\bar{b}+\bar{c}$ is equally inclined to $\bar{a},\bar{b},\bar{c}$

1) Both I & II are true 2) Both I & II are false

3) I is false, II is ture 4) I is true, II is false

2. **I: If the vectors $\bar{a}$ = (1,x,-2), $\bar{b}$ = (x,3,-4) are mutually perpendicular ,then x = 2**

II: If $\bar{a}=\bar{i}+2\bar{j}+3\bar{k},\ \bar{b}=-\bar{i}+2\bar{j}+\bar{k},\ \bar{c}=3\bar{i}+\bar{j}$ and $\bar{a}+t\bar{b}$ is perpendicular to then t=5

1) Only I is true 2) Only II is true

3) both I&II are true 4) neither I nor II are true

3. **Arrange the value of $\left|\bar{a}+\bar{b}+\bar{c}\right|$ in ascending**

order

A: If are mutually perpendicular unit vectors

B: If are vectors of lengths 2,3,4 respectively are perpendicular to respectively

C: If are vectors of lengths 4,4,5 respectively and are perpendicular to respectively.

1) A,B,C 2) C,B,A
3) B,C,A 4) B,A,C

4. **A: Angle between** and is acute angle

$\Rightarrow m \notin \left[-2, -\frac{1}{2}\right]$

R: If $(\bar{a}, \bar{b})$ **is acute then** $\bar{a}.\bar{b} > 0$

1) Both A and R are true and R is the correct explanation of A
2) Both A and R are true and R is not correct explanation of A
3) A is true but R is false
4) A is false but R is true

5. **Arrange the following angles in descending order**

A: If $|\bar{a} + \bar{b}| = |\bar{a} - \bar{b}|$ **then** $(\bar{a}, \bar{b})$

B: If $\bar{a}, \bar{b}, \bar{a} + \bar{b}$ **are unit vectors, then** $(\bar{a}, \bar{b})$

C: If $\bar{a}, \bar{b}, \bar{a} - \bar{b}$ **are unit vectors then** $(\bar{a}, \bar{b})$

1) A,B,C 2) C,B,A
3) B,C,A 4) B,A,C

LEVEL-IV-KEY

01) 1 02) 2 03) 1 04) 1 05) 4

LEVEL-IV-HINTS

1. and

2. (I) , (II)

3. (A)

(B)

(C)

4. does not lies between $\left(-2, \frac{-1}{2}\right) \Rightarrow m \notin \left[-2, \frac{-1}{2}\right]$

5. A) $\bar{a}.\bar{b} = 0 \Rightarrow (\bar{a}, \bar{b}) = \frac{1}{2}$

B) $|\bar{a}| = |\bar{b}| = |\bar{a} + \bar{b}| = 1$

$\Rightarrow \bar{a}^2 + \bar{b}^2 + 2|\bar{a}||\bar{b}|\cos(\bar{a}, \bar{b}) = 1$

$\Rightarrow \cos(\bar{a}, \bar{b}) = \frac{-1}{2}$

C) $|\bar{a}| = |\bar{b}| = |a - b| = 1$

$\Rightarrow |\bar{a}|^2 + |\bar{b}|^2 - 2|\bar{a}||\bar{b}|\cos(\bar{a}, \bar{b}) = \frac{1}{2}$